

Resonant Chemistry

Foundations, Control Solvers, Shell Dynamics, and Relational Candidates

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Scope, Notation, and Method

0.1 Purpose of this textbook

Resonant Chemistry is developed here as a layered research programme rather than as a replacement name for ordinary quantum chemistry. The first obligation of every new layer is to recover a conventional physical control calculation with enough provenance that a later relational hypothesis can be tested against it. The book therefore begins with one- and two-electron control problems, expands to shell-resolved Hartree–Fock, angular multiplets and configuration interaction, and only then introduces shell-level topological and conformational diagnostics.

The primary comparison object of the topological extension is an electronic shell or subshell, not an element label. An atom is a composite system containing a nucleus and a many-electron state; a molecule is a multicentre extension of the same control problem. The text consequently distinguishes

nuclear centres $\oplus$ electronic shells $\oplus$ many-body couplings $\oplus$ candidate relational diagnostics. (1)

No knot is identified with an atom, and no topological correction is inserted into a Hamiltonian merely because a visual analogy is suggestive.

0.2 Epistemic states

Every substantive statement is classified by the strongest status actually supported by the corresponding derivation or receipt:

Established control. Standard physical or mathematical structure used as a baseline.

Model-defined. A quantity defined by the Resonant-Chemistry/TIR formalism but not thereby established as a new physical law.

Candidate. A proposed relation with an explicit validation gate.

Implemented. Executable code exists for the declared model.

Tested. The implementation has passed the stated software or numerical regression tests.

Validated control. A conventional calculation has passed a frozen acceptance criterion and, where applicable, post-hoc external comparison.

Promoted canon. A project-level promotion has been made explicitly; absence of promotion means the result remains at its lower status.

The implications are one-way: implementation is not validation, numerical agreement is not ontology, and a historical receipt is not evidence that a runtime is live now.

0.3 Atomic units and symbols

Unless stated otherwise, atomic calculations use $\hbar = m_e = e = 4\pi\epsilon_0 = 1$. We use Z for nuclear charge, N_e for electron number, n and l for principal and orbital angular-momentum quantum

numbers, and k for subshell occupation when discussing particle–hole symmetry. The capacity of a subshell with orbital angular momentum l is

$$C_l = 2(2l + 1). \quad (2)$$

The many-electron operators L^2 , S^2 and J^2 have their conventional meanings. F^k denotes radial Slater integrals and ζ an explicitly stated spin–orbit scale.

For the candidate topology layer, $\pi_{\text{exch}} \in \{+1, -1\}$ denotes permutation parity in three dimensions, while w_{iA} and w_{ij} denote winding observables only after an oriented projection to an effective plane has been declared. A conformal coordinate is written

$$\eta = \frac{1}{\Phi - z_\star}, \quad (3)$$

where the current exploratory anchor $z_\star = 1/2$ is a falsifiable model choice rather than an experimentally established constant of atomic structure.

0.4 Control-first computational contract

A numerical result enters the textbook only together with four questions: what equations were solved, what data were available to the solver, what acceptance gate was frozen, and what remains open. The repository follows the corresponding chain

$$\text{formalism} \rightarrow \text{implementation} \rightarrow \text{tests} \rightarrow \text{benchmark} \rightarrow \text{interpretation}. \quad (4)$$

Reference data opened after a blind or reference-isolated run are labelled post-hoc. A failed or null result is retained. In particular, the direct knot-gap versus atomic-gap experiment described later did not beat its null model and is preserved as a negative control rather than tuned into a positive association.

0.5 Code, benchmark and semantic layers

The executable package lives under `reschem/`; tests under `tests/`; frozen or replayable numerical checkpoints under `benchmarks/`; and semantic cards under `semantic_cards/`. The physical layer of a semantic card is derived from control calculations. TIR and affective mappings may consume those immutable observables only after a separately defined map with provenance is supplied; they may not rewrite an energy, occupation, multiplet or convergence receipt.

The monograph is the primary formalization surface of the project. Markdown notes may record working discussion, but an equation, assumption, validation rule or negative result that matters to the scientific architecture must also appear in this LaTeX textbook or be cross-referenced from it.

0.6 Book map

Part I constructs the conventional atomic control ladder. Part II scales that ladder from hydrogen through krypton. Part III resolves shell angular structure and atom-specific spectroscopy. Part IV develops finite active-space and sparse correlation controls. Part V introduces shell-level many-body topology, conformal diagnostics and multicentre extensions. Part VI states the admission contract for a future TIR/PhaseNav relational layer without silently inserting it into the validated control Hamiltonians. The appendices bind equations to code, benchmarks, tests and reproducibility gates.

Part I

Control Physics Before Relational Interpretation

Chapter 1

The Atom: Control Physics Before Resonance

1.1 Atomic identity

The first invariant of chemistry is not a resonance label but proton number. We define an atomic species by

$$\mathfrak{A} = (Z, N, q; \mathcal{H}_A, H_A, \rho_A), \quad (1.1)$$

with

$$A = Z + N, \quad N_e = Z - q. \quad (1.2)$$

The element is fixed by Z ; isotope and ionization state are additional coordinates.

1.2 Electronic state space

For N_e indistinguishable electrons the many-electron state belongs to the antisymmetric fermionic space

$$\mathcal{H}_A = \bigwedge^{N_e} [L^2(\mathbb{R}^3) \otimes \mathbb{C}^2]. \quad (1.3)$$

The orbital language used by chemistry is therefore a representation of quantum states, not a literal planetary-orbit model.

1.3 Hamiltonian control layer

In the clamped-nucleus non-relativistic approximation, using atomic units,

$$H_A = \sum_i \left(-\frac{1}{2} \nabla_i^2 - \frac{Z}{r_i} \right) + \sum_{i < j} \frac{1}{r_{ij}} + H_{\text{corr}}. \quad (1.4)$$

The correction sector names finite-mass, relativistic, spin-orbit and QED terms according to the precision required. Resonant Chemistry does not remove this control structure; a new relational layer must coexist with it and survive comparison against it.

1.4 Hydrogenic benchmark

One-electron atoms provide the first exact test of the implementation. At the non-relativistic infinite-nuclear-mass level,

$$E_n = -13.605693122994 \text{ eV} \frac{Z^2}{n^2}, \quad (1.5)$$

and the characteristic radial scale follows

$$r_n \sim a_0 \frac{n^2}{Z}. \quad (1.6)$$

The independent numerical track solves

$$\left[-\frac{1}{2} \frac{d^2}{dr^2} + \frac{\ell(\ell+1)}{2r^2} - \frac{Z}{r} \right] u(r) = E u(r) \quad (1.7)$$

on a finite radial grid. Agreement with the analytic spectrum is therefore a convergence test of the implementation rather than a repeated evaluation of the same formula.

1.5 Entry point for TIR

TIR begins from ordered relations. Its quark/color construction uses holonomic link operators, but the colour $SU(3)$ representation is not copied onto atomic bonding. Instead we reserve an active electronic subspace $\mathcal{E}_A \subset \mathcal{H}_A$. The next stage will construct

$$W_{AB} : \mathcal{E}_B \rightarrow \mathcal{E}_A \quad (1.8)$$

from the relation between electronic subspaces of distinct atoms.

The information quantum

$$\kappa = \frac{\ln 2}{24\pi} \quad (1.9)$$

is carried as TIR metadata in the present atom layer. It does not modify the benchmark spectrum in version 0.1. This separation is deliberate: if a later relational correction is physical, it must produce a discriminating observable rather than merely restating an already-known energy.

1.6 Implementation contract

The accompanying solver reconstructs atomic identity and electron configuration through krypton, evaluates the analytic one-electron benchmark, and independently reconstructs it with a finite-difference radial eigensolver. Multi-electron total energies remain explicitly open until a Hartree–Fock/DFT/CI layer is implemented and validated.

Chapter 2

Helium: the First Interacting Atom

2.1 Why helium is the first chemistry gate

Hydrogen tests the one-electron Hamiltonian, but helium introduces the first unavoidable electron–electron relation. In atomic units, for a two-electron ion of nuclear charge Z ,

$$H = \sum_{i=1}^2 \left(-\frac{1}{2} \nabla_i^2 - \frac{Z}{r_i} \right) + \frac{1}{r_{12}}. \quad (2.1)$$

The $1/r_{12}$ term prevents exact separation into two independent hydrogenic problems. Any proposed Resonant-Chemistry extension must therefore first survive this ordinary interacting benchmark.

2.2 Minimal variational control model

Take a normalized hydrogenic $1s$ spatial orbital with a free effective charge ζ and combine two copies with the antisymmetric spin singlet. For this restricted ansatz the expectation value is

$$E(\zeta) = \zeta^2 - 2Z\zeta + \frac{5}{8}\zeta, \quad (2.2)$$

where $5\zeta/8$ is the analytic expectation value of the electron–electron Coulomb term. Minimization gives

$$\zeta_* = Z - \frac{5}{16}, \quad E_* = -\left(Z - \frac{5}{16}\right)^2. \quad (2.3)$$

For neutral helium, $Z = 2$, hence

$$\zeta_* = \frac{27}{16} = 1.6875, \quad E_* = -2.84765625 \text{ Ha} = -77.4886741 \text{ eV}. \quad (2.4)$$

At the optimum the virial residual $2T + V$ vanishes exactly for this one-parameter family.

2.3 What the error means

The nonrelativistic infinite-nuclear-mass reference energy used only for diagnostics is approximately -2.903724377034 Ha, while the restricted Hartree–Fock limit is approximately -2.861679995612 Ha. The simple shared-orbital ansatz therefore lies about 0.4901% above the Hartree–Fock limit and 1.9309% above the exact nonrelativistic reference. These reference numbers are never used to choose ζ ; the optimum follows analytically from the Hamiltonian.

The gap is scientifically useful. It separates three layers that must not be conflated: orbital relaxation, mean-field exchange, and genuine electron correlation. Resonant Chemistry will not label this missing binding as a TIR effect. The next implementation stage is a numerical restricted Hartree–Fock solver; only after that control layer is reproduced may a relational invariant be tested for additional predictive content.

2.4 TIR status

At this stage $\kappa = \ln 2/(24\pi)$ remains provenance metadata and the future relation operator is inactive. The two-electron benchmark is deliberately conventional. The TIR bridge will enter only through a separately defined relation observable whose effect can be compared against Hartree–Fock and correlated calculations without refitting known helium energies.

Chapter 3

Helium Restricted Hartree–Fock Control

3.1 Self-consistent field equation

The one-parameter $1s^2$ ansatz establishes the first interacting variational bound but does not permit the orbital to relax freely. The next control layer solves the spherical restricted Hartree–Fock equation numerically. For a helium-like two-electron ion with opposite spins sharing one spatial orbital ϕ , the Fock equation is

$$\left[-\frac{1}{2}\nabla^2 - \frac{Z}{r} + J[\phi](r) \right] \phi(r) = \varepsilon\phi(r), \quad (3.1)$$

where the self-interaction is cancelled by exchange and the remaining Coulomb field is that of the opposite-spin electron.

Writing the s orbital as $\phi(r) = u(r)/(r\sqrt{4\pi})$ with $\int_0^\infty |u(r)|^2 dr = 1$, spherical symmetry reduces the Coulomb potential to

$$J(r) = \frac{1}{r} \int_0^r |u(r')|^2 dr' + \int_r^\infty \frac{|u(r')|^2}{r'} dr'. \quad (3.2)$$

The total RHF energy is evaluated as

$$E_{\text{RHF}} = 2\langle\phi|h|\phi\rangle + J, \quad (3.3)$$

with $h = -\nabla^2/2 - Z/r$.

3.2 Discretization and SCF

The implementation uses a uniform radial grid, a second-order three-point finite-difference kinetic operator, direct spherical Coulomb quadrature, density/orbital mixing, and repeated diagonalization of the one-orbital Fock operator until the total energy changes by less than the requested tolerance. No experimental energy is used inside the SCF loop.

Because the finite-difference error is second order in the grid spacing, two nested grids with steps h and $h/2$ are combined by Richardson extrapolation,

$$E_\infty \approx \frac{4E_{h/2} - E_h}{3}. \quad (3.4)$$

3.3 Helium benchmark

For $Z = 2$, $r_{\max} = 20a_0$, and nested grids of 399 and 799 interior points, the implementation obtains

$$E_{399} = -2.8534029917 \text{ Ha}, \quad (3.5)$$

$$E_{799} = -2.8596016563 \text{ Ha}, \quad (3.6)$$

$$E_{\text{Richardson}} = -2.8616678778 \text{ Ha}. \quad (3.7)$$

Against the diagnostic Hartree–Fock limit $-2.861679995612 \text{ Ha}$, the relative discrepancy is approximately 4.23×10^{-6} , or 0.000423%. Both SCF calculations converge in 19 iterations under the default high-accuracy gate.

3.4 Interpretation and TIR boundary

This result closes the first mean-field control requirement for the atom programme: the code can recover the conventional helium RHF limit without any TIR correction. Therefore the missing correlation energy must not be relabelled as “resonance” or as an informational interaction. The next control layer is an explicitly correlated or configuration-interaction calculation. Only after conventional correlation is quantified will a TIR relational observable be introduced, and it must predict an independent invariant or residual rather than repair a known energy by fitting.

Chapter 4

Helium Beyond Hartree–Fock: A Controlled Correlation Layer

4.1 Why helium is the first correlation gate

For the two-electron helium atom, restricted Hartree–Fock (RHF) already solves the self-consistent one-particle problem to high accuracy, but its single-determinant form cannot represent the full correlated dependence of the electronic state. In the conventional definition, the correlation energy is the difference between the exact non-relativistic energy and the Hartree–Fock limit,

$$E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}}. \quad (4.1)$$

This makes helium an ideal control problem: the missing structure is quantitatively known, while the system remains small enough to expose every approximation.

4.2 Finite radial CI basis

The present ResChem control solver first converges a spherical RHF orbital on a radial grid. The final Fock operator is then diagonalized to obtain a finite set of orthonormal radial s orbitals

$$\{u_p(r)\}_{p=1}^M. \quad (4.2)$$

From these orbitals we construct one-electron matrix elements h_{pq} and Coulomb integrals

$$(pq|rs) = \iint \phi_p(1)\phi_q(1)\frac{1}{r_{12}}\phi_r(2)\phi_s(2) d1 d2. \quad (4.3)$$

For spherical s orbitals, angular integration reduces the Coulomb kernel to

$$\frac{1}{r_{>}} = \frac{1}{\max(r_1, r_2)}, \quad (4.4)$$

so the remaining integral is evaluated directly on the radial grid.

4.3 Exact diagonalization inside the finite basis

Each spatial orbital is paired with spin α and β , and all two-electron Slater determinants in the resulting spin-orbital space are generated. The second-quantized Hamiltonian

$$\hat{H} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} (pq|rs) a_p^\dagger a_r^\dagger a_s a_q \quad (4.5)$$

is then diagonalized exactly within that finite orbital space. Thus the term “CI” here means exact configuration interaction in the supplied radial s -orbital basis, not an exact solution of the full helium problem.

4.4 What this layer captures and what it cannot

The radial CI state permits the two electrons to occupy and mix multiple radial configurations. Consequently its energy must lie below the RHF energy computed on the same numerical grid. However, an s -only basis cannot reproduce angular correlation. The residual gap to the exact non-relativistic helium energy is therefore expected and is itself a diagnostic rather than a failure.

With five spatial s orbitals and Richardson extrapolation from 399 and 799 radial points, the current control implementation gives approximately

$$E_{\text{CI},s} \approx -2.862747725 E_h. \quad (4.6)$$

Relative to the Hartree–Fock limit $E_{\text{HF}} \approx -2.861679996 E_h$, the finite radial CI lowers the energy by approximately

$$\Delta E_{\text{radial corr}} \approx 1.068 \times 10^{-3} E_h, \quad (4.7)$$

which is about 2.54% of the full non-relativistic correlation gap to $E_{\text{exact}} \approx -2.903724377 E_h$.

4.5 ResChem/TIR interpretation gate

No TIR correction is fitted to these energies. Instead, the correlated control state supplies new state descriptors that may later feed a relational observable. The first admissible TIR candidate must therefore satisfy all of the following:

1. it is computed from the state or relations between state subspaces, not from the known target energy;
2. it has a fixed transformation law and explicit provenance;
3. it predicts or organizes an observable without refitting the helium reference value;
4. failure of the prediction demotes or rejects the candidate.

This separation is central to the Resonant Chemistry programme: conventional correlation is established first, and only then is a relational interpretation allowed to enter.

Chapter 5

State Invariants and Semantic Entity Cards

5.1 From labels to state-derived semantics

A semantic card must not be a list of human associations attached to an element or molecule name. Its physical backbone should be generated from state variables and invariants, with every derived field tied to explicit repository evidence. For the correlated helium control state, the first such object is the spin-summed one-particle reduced density matrix

$$\gamma_{ij} = \sum_{\sigma} \langle \Psi | a_{i\sigma}^{\dagger} a_{j\sigma} | \Psi \rangle. \quad (5.1)$$

For a two-electron state,

$$\text{Tr } \gamma = 2. \quad (5.2)$$

5.2 Natural occupations

The eigenvalues of γ are the natural occupation numbers n_k . Because eigenvalues are unchanged by unitary rotations of the one-particle basis, the ordered spectrum

$$\mathbf{n} = (n_1, n_2, \dots) \quad (5.3)$$

provides a basis-independent descriptor within the represented active space.

For the present five-orbital radial CI helium state on the 799-point grid, the leading occupations are approximately

$$\mathbf{n} \approx (1.999012068, 9.87928 \times 10^{-4}, 3.93 \times 10^{-9}, \dots). \quad (5.4)$$

The departure from $(2, 0, 0, \dots)$ is a direct signature that the state is not a single closed-shell determinant within this finite model.

5.3 One-body entropy

Normalizing the occupations by particle number,

$$p_k = \frac{n_k}{2}, \quad (5.5)$$

we define the one-body spectral entropy

$$S_1 = - \sum_k p_k \ln p_k, \quad (5.6)$$

and the corresponding linear entropy

$$L_1 = 1 - \sum_k p_k^2. \quad (5.7)$$

For the same helium control state,

$$S_1 \approx 0.00425445, \quad L_1 \approx 9.87444 \times 10^{-4}. \quad (5.8)$$

These quantities are not emotions and are not introduced as TIR postulates. They are standard state-derived descriptors that quantify departure from a rank-one normalized one-body spectrum.

5.4 Semantic-card contract

A calculation card separates at least the following layers:

1. identity and stoichiometric data;
2. conventional physical-control properties and machine-readable source artifacts;
3. state invariants and derived descriptors;
4. explicit relations to other cards or deterministic selectors over the card registry;
5. provenance holonomy recording how the entity was generated from parent cards and operations;
6. physical holonomy, which remains `NOT_COMPUTED` unless winding, phase or topology observables have actually been evaluated with provenance;
7. interpretive TIR and affective mappings, which remain unassigned until an explicit rule with provenance is supplied;
8. epistemic state distinguishing input, model-defined object, emergent candidate, tested control, validated result and promoted canon.

This ordering prevents reverse fitting. An affective or relational interpretation may consume the invariant descriptor vector, but it may not rewrite the underlying control observables.

5.5 Cards as calculation objects

The card layer is not only documentation. The current repository exposes an addressable registry in which a downstream calculation can request one or more card identifiers and receive a deterministic context containing their properties, invariants and relation targets. The complete context is canonically serialized and hashed. A calculation can therefore record not only the code and method that ran, but the exact semantic/entity state it consumed.

This gives the calculation boundary the form

$$\mathcal{C} = (\{C_i\}, \mathcal{R}, H_{\text{ctx}}), \quad (5.9)$$

where C_i are entity cards, $\mathcal{R}$ is the selected relation neighbourhood and H_{ctx} is the context hash. Equality of context hashes establishes equality of the serialized card context; it does not establish physical equivalence of two chemical states.

The current registry includes canonical neutral atomic cards through Kr, nondestructive evidence overlays, persisted model/gate cards, deterministic compound-relation candidates, competing relational-state candidates and molecular-screen cards. High-cardinality candidate populations are regenerated from the same frozen scientific generators rather than maintained as manually copied tables.

5.6 Relation graph

Relations are first-class records. A relation contains a source card, predicate, target and provenance. The target may be an exact card identifier or a deterministic selector such as a neutral atomic card with a specified element symbol. Typical predicates include

HAS_ATOMIC_COMPONENT, DERIVED_BY_MODEL, HAS_CENTRE, HAS_LIGAND, SCREENED_STATE_CANDIDATE. (5.10)

These graph edges organize computation and provenance. They are not automatically identified with the TIR operator W_{ij} ; that operator remains reserved until a separately defined physical/semantic admission rule is supplied.

5.7 Provenance holonomy versus physical holonomy

The term holonomy is used in two deliberately separated senses.

Provenance holonomy. Every generated card records its parent-card identifiers, generating operation and a deterministic lineage hash. This answers the operational question “through which path in model/state space did this entity enter the calculation registry?” For example, a v0.13 relational state records descent from the closed-shell activation and competing-state ensemble gates, and a v0.14A1 molecular-screen card records descent from the three frozen v0.13 state candidates for the same composition.

Physical holonomy. A physical-holonomy field may contain projected winding, Berry/connection phase, pair winding or another explicitly defined topology observable only when that quantity has actually been computed and its source artifact is named. The existence of a provenance loop, a semantic relation cycle or a topology-family label is not sufficient. Thus

$$\text{provenance holonomy} \not\equiv \text{physical holonomy}. \quad (5.11)$$

This boundary is especially important for the shell-topology diagnostics of later chapters.

5.8 Emergent entities

A model may generate an entity that was not entered manually. Examples include the 231 v0.1 binary relation candidates and the 27 v0.13 competing relational states. Such an object receives an ordinary card with properties, invariants, parents and relations, but its emergence status is

$$\text{MODEL_DEFINED_EMERGENT_CANDIDATE}. \quad (5.12)$$

Registry membership means that the candidate is addressable for calculation; it does not promote the candidate to a physically realized molecule, local minimum, ground state or canonical ontology. Independent admission gates remain mandatory.

The same rule applies to future entities that emerge from graph composition, symmetry reduction, topology diagnostics or other deterministic transformations. A new entity should be carded at the point it becomes an addressable computational object, and its generating parents and operation should remain recoverable.

5.9 Current coverage invariant

The repository now treats semantic-card coverage as a maintained gate. Every newly introduced atomic species, compound model, state-ensemble layer or molecular screening/admission gate must receive a card or overlay in the same development cycle. CI checks that the current H–Kr atom coverage exists, that the compound/model trajectory through v0.14A1 has projections, that dynamic high-cardinality entity counts agree with their frozen generators, and that the molecular card set agrees exactly with the machine-readable expected/completed formulae.

The same gate rejects silent semantic promotion. In particular, the current ArBr_2 state remains missing execution rather than chemical failure, and the v0.14A1 cards cannot claim a Hessian minimum, ground-state ranking or geometry-only electronic topology.

5.10 Entry point for a future TIR map

A future map may be written abstractly as

$$\mathcal{F}_{\text{TIR}} : \mathcal{I}(C, \mathcal{R}) \longrightarrow \mathcal{S}, \quad (5.13)$$

where $\mathcal{I}(C, \mathcal{R})$ is a provenance-preserving vector of card invariants and relations and $\mathcal{S}$ is a semantic coordinate space. The map itself is a candidate model and must have a transformation law, provenance and falsification criterion. The current card registry establishes the physical/-computational domain of such a future map; it does not pre-assign its semantic coordinates.

Chapter 6

Lithium: The First Open-Shell Atom

6.1 Why lithium changes the problem

Lithium is the first atom in the present construction whose neutral ground configuration is open-shell,

$$1s^2 2s^1, \quad (6.1)$$

with three electrons and a doublet- S control target. A closed-shell restricted Hartree–Fock ansatz is therefore no longer appropriate. The control layer must distinguish the two spin sectors while preserving the Pauli principle and same-spin exchange.

6.2 Radial unrestricted Hartree–Fock control

Version 0.1 uses a spherical radial unrestricted Hartree–Fock (UHF) model restricted to s orbitals. The spin occupancies are

$$N_\alpha = 2, \quad N_\beta = 1. \quad (6.2)$$

For each spin sector,

$$F_\sigma = h + \sum_j J_j - \sum_{j \in \sigma} K_j, \quad (6.3)$$

where h contains radial kinetic energy and nuclear attraction, J_j is the Coulomb contribution of occupied orbital j , and K_j is included only for same-spin orbitals.

For spherical s orbitals the Coulomb kernel after angular integration is

$$G(r, r') = \frac{1}{\max(r, r')}. \quad (6.4)$$

The exchange operator remains nonlocal and is therefore represented as a dense radial matrix rather than replaced by a local screening potential.

6.3 Energy functional

For occupied spin orbitals $\{\phi_i\}$,

$$E_{\text{UHF}} = \sum_i \langle i | h | i \rangle + \frac{1}{2} \sum_{ij} [J_{ij} - \delta_{\sigma_i \sigma_j} K_{ij}]. \quad (6.5)$$

Self-Coulomb terms cancel exactly against self-exchange for a given spin orbital.

6.4 Spin diagnostic

An unrestricted determinant need not be an exact eigenstate of $\hat{S}^2$. For $N_\alpha = 2$ and $N_\beta = 1$, the diagnostic used here is

$$\langle S^2 \rangle = S_z(S_z + 1) + N_\beta - \sum_{ij} |\langle \phi_i^\alpha | \phi_j^\beta \rangle|^2. \quad (6.6)$$

The ideal doublet value is $S(S + 1) = 3/4$. On the 801-point control grid, the present solver gives approximately

$$\langle S^2 \rangle \approx 0.7500161, \quad (6.7)$$

so spin contamination is approximately 1.61×10^{-5} .

6.5 Grid convergence

The raw radial discretization converges from above. With $r_{\max} = 30 a_0$ the current implementation gives approximately

$$E_{400} = -7.333144359 E_h, \quad (6.8)$$

$$E_{801} = -7.407328552 E_h. \quad (6.9)$$

Using the same second-order Richardson convention as the helium control,

$$E_R = \frac{4E_{801} - E_{400}}{3} \approx -7.432056616 E_h. \quad (6.10)$$

The external Hartree–Fock reference used only for diagnosis is approximately $-7.432727 E_h$, so the present spherical radial control is within roughly 9.0×10^{-5} relative error. This reference value is not an input to the SCF iteration.

6.6 Virial diagnostic

On the 801-point grid the energy components are approximately

$$T = 7.356885300 E_h, \quad (6.11)$$

$$V_{\text{nuc}} = -17.041160101 E_h, \quad (6.12)$$

$$V_{ee} = 2.276946249 E_h, \quad (6.13)$$

with a finite-grid virial residual

$$2T + V_{\text{nuc}} + V_{ee} \approx -0.0504433 E_h. \quad (6.14)$$

Unlike the extrapolated total energy, this raw-grid residual is not yet extrapolated and remains a convergence diagnostic.

6.7 Semantic-card consequences

Lithium introduces new card features that were absent for helium: an open valence shell, a nonzero spin degree of freedom, a distinction between α and β state sectors, and a measurable spin-purity diagnostic. These quantities are physical control descriptors. No TIR or affective meaning is assigned to them in version 0.1.

Chapter 7

First Blind Mass-Solving Experiment

7.1 Purpose

The first mass-solving experiment tests whether the atom solver can be applied repeatedly without species-specific tuning. The benchmark is deliberately restricted to atomic species whose occupied orbitals belong to the spherical s sector, so that the current radial Hartree–Fock geometry is not asked to represent missing p -shell angular structure.

The frozen blind domain is

$$1 \leq N_e \leq 4, \quad N_e \leq Z \leq 4, \quad (7.1)$$

which generates ten species: H, He⁺, Li²⁺, Be³⁺, He, Li⁺, Be²⁺, Li, Be⁺ and Be.

7.2 Blind protocol

No external reference energy is supplied to the solver and no TIR correction enters the Hamiltonian. Each species is solved using unrestricted Hartree–Fock on a uniform radial grid. The radial box is scaled before the run as

$$r_{\max} = 30 \frac{N_e}{Z} a_0, \quad (7.2)$$

which expresses the expected compactification of positively charged isoelectronic ions without using a fitted energy. Calculations use 400 and 801 interior grid points followed by second-order Richardson extrapolation,

$$E_{\infty} \approx \frac{4E_{h/2} - E_h}{3}. \quad (7.3)$$

7.3 Result

All ten systems converged without manual rescue. The four one-electron ions recover the exact hydrogenic law $E = -Z^2/2$ with a maximum relative error of approximately 9.75×10^{-7} . For the neutral sequence, the extrapolated energies are

$$E_{\text{H}} = -0.499999512660 E_h, \quad (7.4)$$

$$E_{\text{He}} = -2.861619787634 E_h, \quad (7.5)$$

$$E_{\text{Li}} = -7.432056591794 E_h, \quad (7.6)$$

$$E_{\text{Be}} = -14.569175169395 E_h. \quad (7.7)$$

Only after this blind run were independent Hartree–Fock limits opened. The resulting relative errors for H, He, Li and Be are approximately 9.75×10^{-7} , 2.10×10^{-5} , 9.02×10^{-5} and 2.64×10^{-4} , respectively. The increasing error is accompanied by a growing virial residual and therefore identifies radial discretization and the nuclear cusp as the next numerical bottleneck.

7.4 Interpretation

This is a successful architectural gate but not a claim of universal atomic accuracy. The solver demonstrates repeatability, spin-sector stability and systematic error structure over the complete present s -shell domain. It also produces a falsification signal: extending the same representation directly to boron would be invalid because the $2p$ shell introduces angular structure absent from the present Hamiltonian representation.

The next solver revision should therefore introduce an adaptive or logarithmic radial mesh and then add explicit $l = 1$ channels before any blind batch containing B–Ne is attempted. TIR and affective semantics remain outside the energy functional during these control tests.

Part II

Scalable Atomic Control from Hydrogen to Krypton

Chapter 8

Multichannel Hartree–Fock and Periodic Coverage

8.1 Multi-channel atomic Hartree–Fock control: from configuration to s , p , and d channels

The first atom solver chapters deliberately started with hydrogenic and s -shell systems. The present layer removes that artificial boundary. The electronic configuration is no longer duplicated inside a solver: it is consumed directly from the canonical atom constructor, which supplies occupied subshells (n, l, q) through krypton.

For each radial channel the one-electron operator is

$$h_l = -\frac{1}{2} \frac{d^2}{dr^2} + \frac{l(l+1)}{2r^2} - \frac{Z}{r}. \quad (8.1)$$

The centrifugal term therefore enters because the configuration requests a nonzero l , rather than through an element-specific branch.

8.1.1 Spherical direct field and nonlocal exchange

The radial charge density is

$$P(r) = \sum_a q_a u_a(r)^2, \quad (8.2)$$

with $\int u_a^2 dr = 1$. Its spherical Coulomb potential is

$$V_J(r) = \frac{1}{r} \int_0^r P(r') dr' + \int_r^\infty \frac{P(r')}{r'} dr'. \quad (8.3)$$

Exchange is retained as a nonlocal multipole operator rather than replaced by a fitted local correction. For target angular momentum l_a and source l_b , allowed multipoles carry the exact angular coefficient

$$c_k(l_a, l_b) = \begin{pmatrix} l_a & l_b & k \\ 0 & 0 & 0 \end{pmatrix}^2, \quad (8.4)$$

and the radial kernel is

$$G_k(r, r') = \frac{r_{\leq}^k}{r_{>}^{k+1}}. \quad (8.5)$$

In v0.1 all coefficients required by $l \leq 2$ are frozen explicitly, so the runtime remains NumPy/SciPy-only.

Open subshells follow Hund occupancy counts for α and β electrons, but magnetic substates are spherically averaged. Consequently this is an *average-of-configuration* Hartree–Fock control model. Closed shells are directly comparable to spherical atomic HF; open p and d shells are not yet term-resolved multiplet calculations. This distinction is mandatory and is stored in every benchmark and semantic card.

8.1.2 Blind H–Ne checkpoint

No reference energy, TIR correction, or semantic/affective parameter enters the solve. A 26-function even-tempered Slater radial basis and 1800-point logarithmic quadrature produced the following first multi-channel blind checkpoint:

Atom	Configuration	E / E_h	target S
H	$1s^1$	-0.500000000000	1/2
He	$1s^2$	-2.861614881532	0
Li	$1s^2 2s^1$	-7.432610713447	1/2
Be	$1s^2 2s^2$	-14.572750829746	0
B	$1s^2 2s^2 2p^1$	-24.414563685151	1/2
C	$1s^2 2s^2 2p^2$	-37.530502491026	1
N	$1s^2 2s^2 2p^3$	-54.403396984787	3/2
O	$1s^2 2s^2 2p^4$	-74.620738988853	1
F	$1s^2 2s^2 2p^5$	-99.162355082192	1/2
Ne	$1s^2 2s^2 2p^6$	-128.543887472990	0

All ten systems converged. The largest absolute virial residual in this checkpoint is $4.97 \times 10^{-5} E_h$. Hydrogen also provides an internal self-interaction gate: the direct Coulomb term is cancelled by exact one-electron exchange to numerical precision.

8.1.3 Why this is not yet H–Kr PASS

The same machinery already consumes $3s$, $3p$, $4s$, $3d$, and $4p$ configurations from the atom constructor. Prototype runs through the third period mostly converge, but aluminium and selected open- $3d$ cases exhibit a two-cycle SCF oscillation. No element-specific damping constant is admitted as a fix. Promotion beyond the H–Ne checkpoint is blocked until a single global SCF stabilizer resolves those cases.

This is an important methodological constraint: failure of a numerical fixed-point iteration is not repaired by changing the physical model or by inserting a semantic correction. TIR remains absent until the conventional control layer is reproducible.

8.1.4 Semantic-card output

For every atomic species the solver exports quantities suitable for later relational analysis without assigning meanings to them: total energy, virial residual, target spin, and for each occupied spin channel the mean radius and radial width. These are physical state coordinates. The semantic and affective layers remain explicitly unassigned and may later consume these coordinates, but may not rewrite them.

8.2 Third Period: Global SCF Stabilization from Na to Ar

The first multi- ℓ implementation reproduced the second period but exposed a two-cycle SCF instability in aluminium. The failure was numerical rather than element-specific: the direct and exchange fields alternated between two nearby density configurations while the total energy remained close to stationary.

The correction introduced here acts only on the fixed-point iteration. For each angular/spin block we form the generalized commutator residual

$$R = FPS - SPF,$$

where F is the raw Fock matrix, P the occupied radial density matrix and S the non-orthogonal Slater-basis overlap matrix. A Pulay/DIIS extrapolation is then constructed from a finite history of (F, R) pairs. No atomic reference energies, TIR corrections or element-specific branches enter the iteration.

8.2.1 Historical single-stage checkpoint

A frozen v0.1 checkpoint used one parameter set for every neutral atom from sodium through argon,

$$\text{basis} = 15, \quad N_r = 600, \quad \lambda_{\text{damp}} = 0.25,$$

with DIIS beginning at iteration three. In the environment in which the benchmark was produced, all eight systems converged and the recorded control energies were

$$\begin{aligned} E_{\text{Na}} &= -161.8214667868 E_h, \\ E_{\text{Mg}} &= -199.5688114017 E_h, \\ E_{\text{Al}} &= -241.7481654795 E_h, \\ E_{\text{Si}} &= -288.6907148139 E_h, \\ E_{\text{P}} &= -340.6375397417 E_h, \\ E_{\text{S}} &= -397.2889482732 E_h, \\ E_{\text{Cl}} &= -459.2240885149 E_h, \\ E_{\text{Ar}} &= -526.6829149879 E_h. \end{aligned}$$

The Hund-spin sequence of the $3p$ shell emerges from the configuration generator as

$$\frac{1}{2}, 1, \frac{3}{2}, 1, \frac{1}{2}, 0$$

for Al through Ar. The semantic-card layer stores this spin coordinate together with the outer-channel mean radius, radial width, energy and virial residual. These are control observables only; TIR semantics and affective coordinates remain unassigned.

8.2.2 Reproducibility correction: the robust ladder is normative

A later full-suite CI run on Python 3.12.13 with newer NumPy/SciPy reproduced 101 tests but showed that the historical fixed single-stage parameters can leave aluminium just outside the convergence condition. This does not license changing the aluminium physics or adding an element-specific damping constant. Instead it confirms that a marginal numerical fixed-point recipe should not be the release contract.

The normative solver entry point therefore uses a global escalation ladder. An atom advances to a larger basis and more conservative DIIS parameters only when either SCF convergence fails

or the virial quality gate fails. The escalation condition depends on numerical quality, never on the element identity. The current regression test applies the same global ladder to every atom from Na through Ar.

This becomes essential in the $3d$ region, where false stationary points can satisfy an energy-only stopping rule while violating the virial check strongly. A useful negative result is that a later numerical stage is not universally superior: if an earlier stage already satisfies the frozen physical quality gate, the robust algorithm accepts it rather than blindly maximizing basis size and iteration depth.

Current boundary. Na–Ar remains a tested control interval under the global non-element-specific quality ladder. The historical fixed-stage energies remain a frozen benchmark rather than the cross-version convergence contract. TIR remains absent from the fixed-point stabilization, and a numerical-library change that alters convergence must be recorded rather than hidden by loosening the scientific interpretation.

8.3 Control Coverage from Hydrogen to Krypton

The atom-control layer now spans every neutral element from hydrogen through krypton, $1 \leq Z \leq 36$. This statement is a coverage claim, not a claim of exact many-body spectroscopy. The purpose of the layer is to provide a reproducible, unfitted physical state vector before any TIR or semantic interpretation is permitted to enter.

8.3.1 Architecture

The calculation is factored into four layers:

1. `atom.py` constructs the deterministic electronic configuration through Kr, including the neutral Cr and Cu exceptions;
2. `atomic_hf_average.py` supplies the spherical average-of-configuration Hartree–Fock operator with s , p , and d angular channels and nonlocal multipole exchange;
3. `atomic_hf_diis.py` stabilizes the SCF fixed-point iteration by Pulay/DIIS and applies a quality ladder independent of element identity;
4. `atomic_mass_h_to_kr.py` exposes batch solving for an arbitrary range and for the complete H–Kr interval.

The robust acceptance condition is

$$\text{PASS} \iff \text{SCF converged} \wedge E \in \mathbb{R} \wedge |2T + V| < 2 E_h. \quad (8.6)$$

The $2E_h$ threshold is intentionally a coarse control gate. It is not a high-precision virial criterion and therefore does not justify promotion to benchmark-quality atomic HF.

8.3.2 Blind checkpoint sequence

The coverage was accumulated in three independent checkpoints:

$$\text{H} \rightarrow \text{Ne} \quad \rightarrow \quad \text{Na} \rightarrow \text{Ar} \quad \rightarrow \quad \text{K} \rightarrow \text{Kr}. \quad (8.7)$$

Reference energies were not available to the solver during these runs. TIR corrections and affective mappings were absent by construction.

The final K–Kr batch contains 18 species and all 18 satisfy the frozen control quality gate. The convergence ladder is genuinely adaptive: most elements pass the first stage, Co and Cu

require a later stage, Zn requires the third stage after a false stationary point is rejected by the virial test, and the Ga–Kr tail is verified on the fourth global fallback. No branch is conditioned on the element name or atomic number beyond the physical configuration itself.

A useful negative result is that the largest numerical stage is not universally superior. Vanadium passes an earlier stage with a small virial residual, while the same stage-4 parameter set can worsen its virial quality. The robust algorithm must therefore accept the first physically admissible fixed point rather than blindly maximize basis size and iteration depth.

8.3.3 Post-hoc krypton check

After the blind calculation was frozen, the krypton result was compared with the Roothaan–Hartree–Fock compilation of Bunge and collaborators. The control solver gives

$$E_{\text{Kr}}^{\text{control}} = -2751.8469168706 E_h, \quad (8.8)$$

while the published RHF value is approximately

$$E_{\text{Kr}}^{\text{RHF}} = -2752.054969 E_h. \quad (8.9)$$

The relative difference is about 7.56×10^{-5} . This validates the usefulness of the common control solver while leaving a clear distinction between control coverage and high-precision atomic HF.

8.3.4 Semantic contract

Every newly covered species has a semantic card. The physical layer may contain configuration, energy, spin target, virial residual, radial information, and solver provenance. The TIR semantic axes remain `CANDIDATE_UNASSIGNED`; the relation operator W_{ij} remains reserved; and affective mappings remain `RESERVED_UNASSIGNED`. A future semantic or affective mapping may consume the immutable physical invariants but may not rewrite them.

8.3.5 Remaining atomic work

The H–Kr control layer is therefore complete with caveats, but the atomic programme is not finished. The next atomic refinements are: tighter treatment of the low-margin Co solution, term-resolved open-shell multiplets, systematic comparison against external HF references, correlation beyond HF, and only then the first TIR relational observable that is not fitted back into the control energy.

Part III

Shell Angular Structure and Spectroscopic Control

Chapter 9

Equivalent-Shell Algebra, Spin–Orbit Ordering, and Radial Spectroscopy

9.1 Angular Multiplets: From Configuration Average to LS Structure

The H–Kr control solver establishes a common radial/electrostatic baseline, but an open shell is not fully specified by a configuration label such as $2p^2$ or $3d^7$. The next layer resolves the allowed electrostatic LS multiplets without modifying the already verified control energies.

9.1.1 Microstate construction

For an equivalent-electron shell with orbital angular momentum l , the one-electron spin-orbital set is

$$\{|lm\sigma\rangle : m = -l, \dots, l; \sigma = \pm \frac{1}{2}\}.$$

For N electrons the antisymmetric determinant basis therefore contains

$$\binom{2(2l+1)}{N}$$

microstates. The implementation constructs the two-electron Coulomb operator directly in this determinant basis. Angular Coulomb matrix elements are evaluated from the spherical multipole expansion and integer Wigner $3j$ symbols computed by the Racah factorial formula. Radial Slater parameters F^k remain explicit coefficients.

The many-electron operators L^2 and S^2 are constructed independently from second-quantized one-body ladder operators. Each degenerate electrostatic eigenspace is subsequently diagonalized with respect to L^2 and S^2 . Therefore the term label ^{2S+1}L is an output of the angular algebra rather than a hard-coded periodic-table label.

9.1.2 Equivalent p shells

With the dimensionless diagnostic choice $F^0 = 0$, $F^2 = 1$, the solver obtains

$$p^1 \rightarrow {}^2P, \quad p^2 \rightarrow {}^3P, \quad p^3 \rightarrow {}^4S, \quad p^4 \rightarrow {}^3P, \quad p^5 \rightarrow {}^2P, \quad p^6 \rightarrow {}^1S.$$

For p^2 , all 15 microstates are resolved into

$${}^3P(9), \quad {}^1D(5), \quad {}^1S(1),$$

where parentheses give degeneracy. For p^3 , all 20 microstates become

$$^4S(4), \quad ^2D(10), \quad ^2P(6).$$

The same equivalent-shell algebra applies to both the $2p$ sequence B–Ne and the $4p$ sequence Ga–Kr. Spin–orbit splitting into J levels is intentionally outside this layer.

9.1.3 Equivalent d shells

For a dimensionless angular diagnostic $F^0 = 0$, $F^2 = 1$, $F^4/F^2 = 0.625$, the ground-term sequence is

$$^2D, ^3F, ^4F, ^5D, ^6S, ^5D, ^4F, ^3F, ^2D, ^1S$$

for d^1 through d^{10} . In particular, d^5 contains $\binom{10}{5} = 252$ microstates and its lowest angular term is 6S .

This is an angular control result. The ratio $F^4/F^2 = 0.625$ is not fitted to any atom-specific spectrum; atom-specific term splittings require coupling these angular coefficients to radial Slater integrals extracted from the converged atomic state.

9.1.4 Status and next gate

This layer promotes equivalent-shell LS structure from OPEN to IMPLEMENTED/TESTED for p shells and to a tested angular control diagnostic for isolated d shells. It does not yet claim fine-structure J splitting, explicit $3d$ – $4s$ exchange for chromium, or atom-specific multiplet separation in Hartree/eV. Those are the next gates.

TIR and affective mappings remain absent by construction. Multiplet invariants may later become semantic inputs, but they are not allowed to rewrite the control physics.

9.2 Weak Spin–Orbit Ordering and Ground- J Resolution

The electrostatic LS layer classifies open-shell states by L and S , but spectroscopy resolves total angular momentum J . The next control layer adds the one-electron spin–orbit operator

$$H_{\text{SO}} = \zeta \sum_i \mathbf{l}_i \cdot \mathbf{s}_i$$

to the same determinant microstate basis. In version 0.1, ζ is deliberately dimensionless and small compared with the electrostatic Slater scale. The purpose is to resolve the angular ordering of J branches in the LS-coupling regime, not to claim atom-specific fine-structure magnitudes.

The total operator J^2 is built independently from many-electron ladder operators. The lowest eigenspace of $H_{\text{Coulomb}} + H_{\text{SO}}$ is projected onto J^2 , making the ground- J value an output of the same microstate calculation.

For equivalent p shells the calculated sequence is

$$J(p^1 \dots p^6) = \left(\frac{1}{2}, 0, \frac{3}{2}, 2, \frac{3}{2}, 0 \right).$$

Together with the LS sequence this gives the familiar weak-coupling ground branches

$$^2P_{1/2}, ^3P_0, ^4S_{3/2}, ^3P_2, ^2P_{3/2}, ^1S_0.$$

For equivalent d shells the calculated sequence is

$$J(d^1 \dots d^{10}) = \left(\frac{3}{2}, 2, \frac{3}{2}, 0, \frac{5}{2}, 4, \frac{9}{2}, 4, \frac{5}{2}, 0 \right),$$

corresponding to

$$^2D_{3/2}, ^3F_2, ^4F_{3/2}, ^5D_0, ^6S_{5/2}, ^5D_4, ^4F_{9/2}, ^3F_4, ^2D_{5/2}, ^1S_0.$$

This layer reproduces angular branch ordering only. Absolute splittings require an atom-specific spin-orbit constant or an explicit relativistic one-electron operator coupled to the converged radial state. Chromium remains a special configuration-level gate because its neutral ground configuration $3d^54s^1$ requires explicit inter-subshell coupling rather than an isolated equivalent- d shell.

No TIR correction and no affective mapping is present in this control layer.

9.3 Atom-Specific Period-2 Spectroscopy from the Radial State

The previous angular layer resolved the allowed LS terms and the ordering of J branches without assigning an atom-specific energy scale. This chapter closes that gap for the neutral second-period $2p^n$ sequence B–Ne without fitting to observed spectra.

9.3.1 Radial Slater integrals

Let $P_{2p}(r)$ denote the normalized shell-averaged radial probability amplitude obtained from the configuration-average atomic Hartree–Fock control state. When alpha and beta radial functions differ, the one-electron shell density is defined by occupancy averaging,

$$g_{2p}(r) = \frac{n_\alpha P_\alpha(r)^2 + n_\beta P_\beta(r)^2}{n_{2p}}, \quad \int_0^\infty g_{2p}(r) dr = 1. \quad (9.1)$$

The radial Slater integrals are then

$$F^k(2p, 2p) = \int_0^\infty dr_1 \int_0^\infty dr_2 g_{2p}(r_1) g_{2p}(r_2) \frac{r_1^k}{r_2^{k+1}}, \quad (9.2)$$

with $k = 0, 2$ for a p shell. No reference spectrum enters this calculation.

9.3.2 Pauli central-field spin-orbit scale

The first atom-specific spin-orbit control parameter is evaluated in the Pauli central-field approximation,

$$\zeta_{2p} = \frac{\alpha^2}{2} \int_0^\infty g_{2p}(r) \frac{1}{r} \frac{dV_{\text{loc}}}{dr} dr. \quad (9.3)$$

For the spherical nuclear-plus-direct field of the other electrons,

$$\frac{1}{r} \frac{dV_{\text{loc}}}{dr} = \frac{Z - Q_{\text{other}}(r)}{r^3}. \quad (9.4)$$

The active electron’s own shell-averaged density is removed before constructing Q_{other} , preventing an explicit Hartree self-field from entering the control spin-orbit scale. Exchange remains nonlocal and is therefore not silently replaced by an ad hoc local exchange potential.

9.3.3 From radial state to spectrum

The atom-specific shell Hamiltonian is

$$H_{\text{shell}} = H_{\text{Coulomb}}[F^0, F^2] + \zeta_{2p} \sum_i \mathbf{l}_i \cdot \mathbf{s}_i. \quad (9.5)$$

It is diagonalized in the same determinant microstate space used in the angular multiplet solver. The resulting eigenvectors are classified by J^2 and monitored with L^2 and S^2 expectations.

For the blind high-resolution control run (24 radial functions, 1500 radial points), representative values are

Atom	F^2 (Ha)	ζ_{2p} (cm ⁻¹)	Ground LS_J
B	0.142681	9.906	$^2P_{1/2}$
C	0.224879	38.118	3P_0
N	0.297700	97.013	$^4S_{3/2}$
O	0.315929	176.565	3P_2
F	0.371492	309.813	$^2P_{3/2}$
Ne	0.427604	517.194	1S_0

Post-hoc comparison with NIST ASD was opened only after the blind calculation. The B 2P splitting is predicted as 14.859 cm⁻¹ versus 15.287 cm⁻¹; F is 464.719 cm⁻¹ versus 404.141 cm⁻¹. Oxygen gives the correct $^3P_{2,1,0}$ ordering with 177.50 and 264.27 cm⁻¹ above the ground level, compared with 158.265 and 226.977 cm⁻¹. The larger errors of some singlet terms identify missing correlation/configuration interaction rather than a reason to fit F^2 by hand.

Epistemic boundary. This layer is promoted as an atom-specific spectroscopy *control* model, not as high-precision atomic structure. The radial state is nonrelativistic Hartree–Fock; the spin–orbit term is Pauli central-field; correlation, Breit interaction, and Dirac self-consistency remain open. TIR and affective mappings remain absent from the control Hamiltonian.

Part IV

Correlation Architecture and the Frozen-Orbital Boundary

Chapter 10

Active Spaces, Correlated Spectra, and Carbon Channel Diagnostics

10.1 Period-2 active-space correlation without refitting

The atom-specific spectroscopy layer of Sec. 9.3 exposed a controlled failure mode: the ordering of the low $2p^n$ terms was correct and the absolute scale was meaningful, but selected singlet and doublet term centers remained too high. This chapter adds the smallest correlation layer that can address that failure without changing any experimental target, refitting any Slater integral, or introducing a TIR correction.

10.1.1 Frozen core and active space

For neutral B–Ne we freeze the closed $1s^2 2s^2$ core and solve it self-consistently. The active electrons occupy an $l = 1$ radial basis generated from the frozen-core Fock operator. Version 0.1 keeps the two lowest radial p functions, so the spin-orbital active space has

$$N_{\text{so}} = 2 \times (2l + 1) \times 2 = 12 \quad (10.1)$$

spin orbitals. For n active p electrons the determinant dimension is therefore

$$\dim \mathcal{H}_{p^n} = \binom{12}{n}. \quad (10.2)$$

This gives 12, 66, 220, 495, and 792 determinants for B through F.

The active Hamiltonian is

$$H_{\text{act}} = \sum_{pq} h_{pq}^{\text{core}} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} (pq|rs) a_p^\dagger a_q^\dagger a_s a_r, \quad (10.3)$$

where the one-electron operator contains the nuclear field and the frozen-core direct and exchange potentials. The two-electron matrix elements use explicit radial multipoles

$$R^k(ab; cd) = \iint u_a(r_1) \nu_c(r_1) \nu_b(r_2) \nu_d(r_2) \frac{r_1^k}{r_2^{k+1}} dr_1 dr_2, \quad k = 0, 2, \quad (10.4)$$

where $u_a(r) \equiv \nu_a(r)$ is the normalized radial function used by the code. The multipoles are combined with the same angular Wigner algebra used by the equivalent-shell solver. Diagonalization is exact inside this finite active space. Consequently, configurations analogous to $2p^{n-2}3p^2$ are not inserted as hand-written corrections; they arise as determinants in the common CI basis.

10.1.2 Reference isolation

Experimental levels were already known from the preceding post-hoc spectroscopy audit, so this stage is not called blind in the human-information sense. The solver itself remains reference-isolated: no NIST value, empirical term energy, fitted F^k , or fitted spin-orbit constant is consumed by the Hamiltonian. The previous benchmark is frozen before the new CI result is evaluated.

10.1.3 Correlation checkpoint

At basis size 18 and 800 radial grid points, the correlated electrostatic term centers are

Target	previous control / cm^{-1}	active CI / cm^{-1}
C 1D	11883.53	10972.16
C 1S	29651.45	25192.84
N 2D	23521.02	18903.85
N 2P	39202.78	30217.19
O 1D	16731.34	15537.00
O 1S	41693.71	36975.43

All six frozen C/N/O targets move toward the experimental term centers. The largest improvement occurs for the nitrogen 2D center, while both carbon and oxygen singlets improve simultaneously. This is the expected qualitative signature of a missing configuration-interaction channel rather than a simple scale error in F^2 .

10.1.4 Current boundary

The result is promoted only as a control correlation layer. The $1s^22s^2$ core is frozen, only two radial p orbitals are retained, s and d virtual channels are absent, and the spin-orbit operator has not yet been rediagonalized in the enlarged correlated space. No high-precision claim is made. The next controlled extension is therefore a larger active space and correlated spin-orbit rediagonalization, followed by core-valence and relativistic tests.

TIR and affective mappings remain absent from the control Hamiltonian.

10.2 Correlated spin-orbit rediagonalization in the period-2 active space

The active-space CI layer of Sec. 10.1 improves the electrostatic term centers without refitting the radial Slater integrals. The next question is whether the previously derived atom-specific spin-orbit scale can be rediagonalized inside the same correlated space without introducing a new empirical parameter.

10.2.1 Conservative projection

The established period-2 spectroscopy control layer already supplies an atom-specific Pauli central-field constant ζ_{2p} . Version 0.1 reuses that same constant and projects the one-electron operator

$$H_{\text{SO}} = \zeta_{2p} \sum_i \mathbf{l}_i \cdot \mathbf{s}_i \quad (10.5)$$

onto the lowest radial p channel of the two-radial-orbital CI basis. The virtual radial p channel and cross-radial spin-orbit blocks are set to zero. This is deliberately conservative: no virtual-channel ζ is inferred from experiment, and no cross-radial scale is introduced by hand.

The correlated Hamiltonian is therefore

$$H_{\text{corr,SO}} = H_{\text{active CI}} + P_{2p}H_{\text{SO}}P_{2p}, \quad (10.6)$$

with P_{2p} the projector onto the physical lowest radial p channel. The full determinant-space Hamiltonian is rediagonalized and classified by J^2 , with L^2 and S^2 retained as approximate labels.

10.2.2 Frozen benchmark replay

Experimental values were already frozen by the earlier post-hoc NIST audit. They are not consumed by the Hamiltonian. Relative to the previous equivalent-shell fine-structure control, the correlated calculation gives

Target	previous / cm^{-1}	correlated / cm^{-1}	frozen reference / cm^{-1}
B $^2P_{3/2}$	14.859	14.859	15.287
C 3P_1	19.157	18.219	16.417
C 3P_2	57.213	54.410	43.413
O 3P_1	177.496	144.907	158.265
O 3P_0	264.274	215.751	226.977
F $^2P_{1/2}$	464.719	383.480	404.141

B is preserved, while every listed C, O, and F target improves. The strongest reduction in relative error occurs for O 3P_0 and F $^2P_{1/2}$.

10.2.3 Rejected stronger ansatz

A prototype in which the physical $2p$ spin-orbit scale was extended to the virtual radial channel through a radial $\langle r^{-3} \rangle$ scaling produced excessive quenching of several splittings. That ansatz is therefore not promoted. The present projector construction keeps the validated physical $2p$ scale and leaves virtual/cross-radial spin-orbit coupling open for a future derivation.

10.2.4 Verification and boundary

The exact repository workflow on GitHub Actions runs both the active-CI and correlated-spectrum tests. At the current checkpoint all seven tests pass. This validates implementation and numerical regression, not high-precision spectroscopy. Core-valence correlation, additional radial p orbitals, s/d virtual channels, virtual spin-orbit matrix elements, Dirac self-consistency, and Breit terms remain open.

TIR and affective mappings remain absent from the control Hamiltonian.

10.3 Carbon correlation channels: why one truncated CI is not enough

The period-2 correlation programme now has enough controlled variants to ask a more specific question than whether “correlation helps”: which active channels lower which carbon term centers, and when does a truncated active space become unbalanced between states?

10.3.1 Radial p convergence

With the $1s^2 2s^2$ core frozen, the two-radial- p active CI gives

$$E(^1D) - E(^3P) = 10972.16 \text{ cm}^{-1}, \quad E(^1S) - E(^3P) = 25192.84 \text{ cm}^{-1}. \quad (10.7)$$

Expanding the radial p space from two to four functions changes these values to

$$10915.10 \text{ cm}^{-1}, \quad 25366.89 \text{ cm}^{-1}, \quad (10.8)$$

respectively. The 1D center converges smoothly downward, whereas 1S moves slightly upward. The missing 1S physics is therefore not simply more radial p flexibility.

10.3.2 Opening the physical $2s$ channel

A naive frozen- $1s^2$ calculation using orbitals generated in the field of the bare core is rejected: it overcontracts the valence functions and yields large errors in both singlets. A mean-field-consistent variant instead keeps the established closed- $1s^2 2s^2$ radial shapes, removes the active- $2s$ mean field from the one-body Hamiltonian, and restores the $2s/2p$ interaction explicitly in FCI. This gives

$$^1D : 14880.50 \text{ cm}^{-1}, \quad ^1S : 23564.40 \text{ cm}^{-1}. \quad (10.9)$$

The 1S center improves strongly while 1D worsens. Thus the radial- p and explicit- $2s$ channels are not redundant.

10.3.3 Balanced $2s + 2p + 3p$ active space

The next model combines physical $2s$ participation with two radial p functions in a four-electron even-parity FCI. The space contains 14 spin orbitals and 561 even-parity determinants. The resulting term centers are

$$^1D : 10584.51 \text{ cm}^{-1}, \quad ^1S : 16214.14 \text{ cm}^{-1}. \quad (10.10)$$

Relative to the frozen post-hoc experimental targets, the 1D error drops to about 3.84%, but the 1S center is driven too low, with an error of about 25.10%. The combined truncated space therefore overcorrelates 1S relative to 3P .

This result is not promoted as a globally superior carbon model. It is promoted only as a channel diagnostic: different term symmetries respond differently to different correlation channels, and a variationally larger active space can improve one excitation while worsening another when orbital relaxation and complementary angular channels are incomplete.

10.3.4 Next gate

The next controlled extension is an even-parity virtual d channel coupled to the frozen- $1s^2 2s^2$ active- p reference. This is preferable to further p -only growth or empirical rescaling because it introduces a genuinely new angular-correlation mechanism. State-averaged orbital optimization may be considered only after the d -channel contribution has been measured separately.

No TIR or affective term is used anywhere in this diagnostic chain.

10.4 Sparse multichannel carbon CI and the frozen-orbital boundary

The channel diagnostics of Sec. 10.3 show that radial p , explicit $2s$, and virtual d correlation affect the carbon singlets differently. Combining these channels increases the determinant space beyond the point where repeated dense diagonalization is a sensible default. The implementation therefore moves to a sparse symmetry-preserving Hamiltonian while leaving the underlying Coulomb formalism unchanged.

10.4.1 Sparse $2s + 2p + 3p + 3d$ model

With the $1s^2$ core frozen, the first sparse balanced space contains the physical $2s$ orbital, two radial p functions, one radial d function, and four active electrons. Restricting to even parity leaves 5346 determinants. The Hamiltonian contains 217140 nonzero elements and is stored in compressed sparse-row form. Only the low spectrum is requested from a Lanczos/ARPACK solver, while L and S are identified from Casimir expectation values built from sparse one-body angular operators.

The resulting low term centers are

$${}^1D = 10615.05 \text{ cm}^{-1}, \quad {}^1S = 17513.78 \text{ cm}^{-1}. \quad (10.11)$$

The 1D center remains close to the frozen post-hoc reference, with about 4.14% relative error, whereas 1S remains overcorrelated relative to 3P .

10.4.2 Opening the first virtual s channel

The next controlled extension adds one reference-shaped virtual s orbital while retaining the two radial p functions and one d radial function. The active space grows to 26 spin orbitals, 7502 even-parity determinants, and 374814 nonzero Hamiltonian elements. The low spectrum becomes

$${}^1D = 12177.40 \text{ cm}^{-1}, \quad {}^1S = 20155.93 \text{ cm}^{-1}. \quad (10.12)$$

The 1S error falls to about 6.89%, the best result for that center in the present frozen-orbital channel sequence, but the 1D error rises to about 19.47%.

10.4.3 What the channel tension means

These calculations remove the simplest remaining explanation of the carbon discrepancy. The problem is not merely an insufficient number of radial p functions, nor a missing $2s$, $3s$, or $3d$ channel considered in isolation. Different term symmetries prefer different balances of the same frozen orbital set. Increasing the variational CI space can therefore improve one excitation while worsening another because the orbitals themselves were optimized for a different reference state.

Accordingly, no single frozen-orbital model in this sequence is promoted as a high-precision carbon solver. The sparse architecture is promoted as a control implementation, and the channel dependence is promoted as a diagnostic result.

10.4.4 Next gate: state-averaged orbital optimization

The next model increment is state-averaged orbital optimization or MCSCF over a common 3P , 1D , and 1S objective. The objective must be defined internally from state energies and

normalization constraints; experimental term values remain excluded from the optimization. This separates genuine variational orbital relaxation from empirical spectrum fitting.

No TIR or affective term is used in any Hamiltonian in this chapter.

10.5 First State-Averaged Active–External Orbital Relaxation Gate

The frozen-orbital diagnostics of Sec. 10.4 identify a structural limitation rather than a missing isolated channel: the 3P , 1D , and 1S terms do not prefer the same fixed orbital representation. The next admissible step must therefore vary the orbital subspace itself while keeping experimental term energies outside the objective.

10.5.1 Why an internal active-space rotation is insufficient

A full configuration interaction calculation in a fixed active orbital span is invariant under unitary rotations within that complete span. Consequently, rotating two already active radial p functions cannot constitute orbital optimization: it changes the representation but not the variational space. A genuine first orbital-relaxation test must mix an active direction with a direction that was previously external to the CI space.

10.5.2 Three-radial- p reference and two-radial- p active space

Let $\{u_{p0}, u_{p1}, u_{p2}\}$ be three mutually orthonormal radial p functions obtained from the same closed $1s^2 2s^2$ reference field. The first radial is retained and the second active radial is allowed to rotate into the external third direction,

$$\tilde{u}_{p1}(\theta) = \cos \theta u_{p1} + \sin \theta u_{p2}. \quad (10.13)$$

The active p span is therefore

$$\mathcal{A}_p(\theta) = \text{span}\{u_{p0}, \tilde{u}_{p1}(\theta)\}. \quad (10.14)$$

At $\theta = 0$ this reproduces the frozen two-radial- p reference. For $\theta \neq 0$, $\mathcal{A}_p(\theta)$ is a different two-dimensional subspace of the three-dimensional reference span. The excluded orthogonal complement changes accordingly, so this transformation is not a gauge rotation of a fixed full-CI space.

10.5.3 Internal state-average objective

For each θ , the even-parity four-valence-electron Hamiltonian is rebuilt in the physical $2s$ plus two-active- p space, diagonalized, and classified by L^2 and S^2 . Define the equal-weight state-average

$$\overline{E}(\theta) = \frac{1}{3} \left[E(^3P; \theta) + E(^1D; \theta) + E(^1S; \theta) \right]. \quad (10.15)$$

All three quantities in Eq. (10.15) are absolute CI eigenenergies from the same Hamiltonian. Experimental term centers do not enter the objective, the orbital generator, or the stopping rule.

More generally, for non-negative normalized weights w_T ,

$$\overline{E}_w(\theta) = \sum_{T \in \{^3P, ^1D, ^1S\}} w_T E(T; \theta), \quad \sum_T w_T = 1. \quad (10.16)$$

The present gate fixes equal weights before evaluation to avoid post-hoc preference for whichever term improves most.

10.5.4 Variational acceptance rule

Let Θ be a symmetric scan containing $\theta = 0$. The candidate passes the minimal variational gate when

$$\min_{\theta \in \Theta} \overline{E}(\theta) \leq \overline{E}(0) + \varepsilon_{\text{num}}, \quad (10.17)$$

with finite classified 3P , 1D , and 1S states throughout the accepted scan. A strict decrease demonstrates that the frozen active subspace was not stationary against the tested external p direction. Equality within numerical tolerance is a valid negative result: it would mean that this particular external direction does not supply the missing relaxation.

10.5.5 Production numerical gate

The production benchmark used 18 radial basis functions, 800 radial grid points, a 10^{-9} Ha SCF tolerance, a 21-point coarse scan over $\theta \in [-1.45, 1.45]$, and an 11-point refinement over the preselected neighboring coarse interval $[-0.435, -0.145]$. The equal $^3P/{}^1D/{}^1S$ weights were fixed before the run. The complete generated benchmark was recorded by GitHub Actions run 31698054125 and its SHA-256 digest is 4991f89450a47480a8ed71df33799bb9da60451770a9471cc182db7ab288fdbbe.

Quantity	Frozen $\theta = 0$	Best sampled $\theta = -0.290$
$\overline{E}$ (Ha)	-5.243337545	-5.246907956
$E(^3P)$ (Ha)	-5.284039512	-5.287703975
$E(^1D)$ (Ha)	-5.235812345	-5.239533714
$E(^1S)$ (Ha)	-5.210160777	-5.213486180
${}^1D - {}^3P$ (cm^{-1})	10584.64	10572.15
${}^1S - {}^3P$ (cm^{-1})	16214.51	16288.92

Table 10.1: Production one-coordinate state-averaged radial- p relaxation benchmark. Experimental term energies are absent from the objective.

The sampled state-average decreases by

$$\Delta \overline{E} = \overline{E}(0) - \overline{E}(-0.290) = 0.0035704116 \text{ Ha} > 0. \quad (10.18)$$

The refined sampled values around the minimum are -5.246893018 Ha at $\theta = -0.319$, -5.246907956 Ha at $\theta = -0.290$, and -5.246860274 Ha at $\theta = -0.261$. Thus the best sampled point is interior to the refinement interval, rather than an edge artifact of the earlier small scan.

This is a *variational* improvement, not a claim of globally improved spectroscopy. Relative to the frozen point, the 1D gap changes by approximately -12.49 cm^{-1} whereas the 1S gap changes by approximately $+74.42 \text{ cm}^{-1}$. All three *absolute* target-state energies decrease sufficiently to lower the fixed state-average, but the two excitation gaps do not move in the same direction. The distinction is essential: the optimization objective is the internal variational energy, not agreement with experimental term centers.

The production verdict is therefore **PASS_VARIATIONAL_DIRECTION**. It establishes that the frozen two-radial- p active space is not stationary along the tested active-external p direction. It does not yet establish a continuously converged optimum or basis/grid convergence of the optimum angle and energy gain.

10.5.6 Implementation boundary

The reference implementation is `reschem/carbon_state_averaged_orbital_relaxation.py`. It constructs the common closed-reference radial basis, evaluates the valence one-body operator

in the frozen- $1s^2$ field, rebuilds two-body Coulomb integrals after each active-external rotation, and classifies the resulting CI eigenstates.

This is deliberately labelled a *first orbital-relaxation gate*, not full MCSCF or CASSCF. Only one radial- p active-external coordinate is varied; the $1s^2$ core remains frozen; s and d orbital relaxation are not yet simultaneous; and spin-orbit coupling is outside this electrostatic optimization layer. A successful gate therefore justifies expansion to a multidimensional state-averaged orbital optimizer, not promotion to a complete carbon solver.

No TIR, PhaseNav, conformal, knot, or affective term enters the Hamiltonian or objective in this section.

Part V

Shell-Level Many-Body Topology and Conformal Diagnostics

Chapter 11

Shell-Level N-Body Topology with Nuclear Centres

11.1 Status and scope

This chapter introduces a *candidate representation layer*. It does not modify the validated Coulomb Hamiltonian and it does not identify knots with atoms. The comparison object is an electronic shell or subshell, while nuclei remain explicit attractive Coulomb centres.

11.2 Control Hamiltonian

For nuclei $A = 1, \dots, M$ and electrons $i = 1, \dots, N$, the nonrelativistic control Hamiltonian in atomic units is

$$H = \sum_i \frac{\mathbf{p}_i^2}{2} - \sum_{i,A} \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} + \sum_{i < j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + H_{\text{corr}} + H_{\text{SO}} + \dots \quad (11.1)$$

The topology layer is diagnostic by default: it labels histories or sectors and does not introduce a new force term.

11.3 Nuclear attractive centres

Define $\mathcal{A}_A = (Z_A, \mathbf{R}_A)$. Each centre contributes $V_A(\mathbf{r}) = -Z_A/|\mathbf{r} - \mathbf{R}_A|$. In a conservative Hamiltonian flow $\mathcal{A}_A$ is not a dissipative attractor in the strict sense. In reduced SCF or relaxation dynamics, basin/attractor language is admissible only for the reduced map.

11.4 Configuration topology and dimensional boundary

For N identical point particles in three spatial dimensions, the unordered collision-free configuration space has fundamental exchange topology governed by the permutation group S_N . Ordinary braid-group topology is fundamental for effective two-dimensional or otherwise constrained motion. Consequently, this work distinguishes three-dimensional exchange permutation data, projection-dependent winding diagnostics, and genuine braid invariants only after a declared effective-2D constraint.

11.5 Shell particle-hole duality

For orbital angular momentum l , the shell capacity is

$$C_l = 2(2l + 1). \quad (11.2)$$

For occupation k , define $k^* = C_l - k$. The self-dual half-filled shell obeys $k = k^* = 2l + 1$. Thus

$$p^1 \leftrightarrow p^5, \quad p^2 \leftrightarrow p^4, \quad p^3 \leftrightarrow p^3, \quad (11.3)$$

$$d^k \leftrightarrow d^{10-k}, \quad d^5 \leftrightarrow d^5, \quad (11.4)$$

$$f^k \leftrightarrow f^{14-k}, \quad f^7 \leftrightarrow f^7. \quad (11.5)$$

11.6 Projected winding observables

For a declared oriented projection $P : \mathbb{R}^3 \rightarrow \mathbb{R}^2 \simeq \mathbb{C}$, define

$$w_{iA} = \frac{1}{2\pi} \oint d \arg P(\mathbf{r}_i - \mathbf{R}_A), \quad w_{ij} = \frac{1}{2\pi} \oint d \arg P(\mathbf{r}_i - \mathbf{r}_j). \quad (11.6)$$

These are representation observables, not projection-independent invariants of unrestricted 3D quantum dynamics.

11.7 Conformal reduction

Let $\Phi[\gamma] \in \mathbb{C}$ be a declared scalar relational coordinate derived from a shell history. Introduce

$$\zeta = \Phi[\gamma] - z_*, \quad \eta = \frac{1}{\zeta}. \quad (11.7)$$

The current candidate uses $z_* = 1/2$ only as a falsifiable anchor. It must be varied under identifiability controls before any physical interpretation.

11.8 Shell-topology state vector

A minimal diagnostic record is

$$\Sigma_{nlk} = (n, l, k, C_l - k, \pi_{\text{exch}}, \{w_{iA}\}, \{w_{ij}\}, \Gamma_{\text{geom}}, \eta), \quad (11.8)$$

where $\pi_{\text{exch}} \in \{+1, -1\}$ is permutation parity and Γ_{geom} denotes an independently defined geometric phase when available.

11.9 Validation ladder

The required order is: recover standard SCF/CI control data; extract shell topology observables without fitting experimental energies; test $p^k \leftrightarrow p^{6-k}$ internally; test transfer from $2p^k$ to $3p^k$; only then compare frozen shell classes with knot/conformal families. Failure at any stage is retained and does not license retuning of the conformal anchor or shell labels.

11.10 Implementation status

The reference implementation `reschem/shell_nbody_topology.py` provides shell capacity and particle-hole duality, nuclear Coulomb centres, 3D exchange parity, declared-plane electron–nucleus and electron–electron winding diagnostics, and the conformal inverse coordinate. These functions are implementation primitives; physical validation against many-electron trajectories remains open.

Chapter 12

Conformal Knot Diagnostics as a Shell-Level Candidate

12.1 Correction of the comparison object

The exploratory knot study originally compared knot-derived descriptors with atom-level feature vectors. That is not the correct structural level for the intended chemistry programme. Chemical bonding and open-shell structure are mediated by electronic subshells, while the nucleus remains an explicit attractive Coulomb centre. The corrected hypothesis therefore has the form

$$\text{topological/conformal class} \longleftrightarrow \text{subshell occupation class } (n, l, k), \quad (12.1)$$

not “knot equals atom”. Element names may provide examples of a given occupation, but they are not allowed to define the mapping.

12.2 Knot-side conformal construction

Let $V_K(q)$ be the Jones polynomial of a knot K and parameterize the unit circle by $q = e^{i\theta}$. Define

$$\Phi_K(\theta) = V_K(e^{i\theta}), \quad \eta_K(\theta) = \frac{1}{\Phi_K(\theta) - z_\star}. \quad (12.2)$$

For the exploratory sweep $z_\star = 1/2$. A diagnostic feature record is

$$\Sigma_K = (c_K, b_K, |\text{wind}_{z_\star} K|, d_{\min}, N_{z_\star}), \quad (12.3)$$

where c_K is crossing number, b_K Jones breadth, $d_{\min} = \min_\theta |\Phi_K(\theta) - z_\star|$, and $N_{z_\star}$ counts numerically detected roots of $V_K(q) - z_\star$ on $|q| = 1$. Signed winding is retained in provenance; its absolute value is used for mirror-robust diagnostics.

The sweep through prime knots up to eight crossings produced the numerical candidate family

$$4_1, 6_3, 8_3, 8_9, 8_{12}, 8_{17}, 8_{18}, \quad (12.4)$$

for which roots of $V_K(q) - 1/2$ were detected on the unit circle within the numerical tolerance of the experiment. This is a numerical observation, not an analytic theorem; exact unit-circle membership remains a symbolic gate.

12.3 The rejected direct spectral match

For each candidate singular knot, the root angles were converted into normalized cyclic gaps. These gap spectra were compared with normalized atomic multiplet gaps from the period-2 control layer. The visually closest first comparison, 8_{18} against oxygen, had empirical null probability of approximately 0.36 under random four-gap partitions. The direct knot-gap versus atom-gap association therefore does not beat a simple null model.

This negative result is retained explicitly:

$$\boxed{\text{direct knot-gap} \leftrightarrow \text{atom-gap matching: NOT PROMOTED.}} \quad (12.5)$$

No relabelling, parameter tuning or choice of a different atom after seeing the null result is admitted as a rescue.

12.4 Shell particle–hole involution

For a subshell of capacity $C_l = 2(2l + 1)$, define the particle–hole involution

$$\mathcal{P}_h : (l, k) \mapsto (l, C_l - k). \quad (12.6)$$

The p shell therefore has the intrinsic pairs

$$p^1 \leftrightarrow p^5, \quad p^2 \leftrightarrow p^4, \quad p^3 \leftrightarrow p^3. \quad (12.7)$$

The middle occupation is self-dual because $k = C_l/2 = 2l + 1$. The same rule yields the self-dual centres d^5 and f^7 .

The equivalent-shell multiplet calculations in Part III already supply a strong control fact: the p^k and p^{6-k} ground-term sequence is paired by occupation independently of whether the radial shell is $2p$, $3p$ or $4p$. This makes shell duality a physically motivated comparison object that exists before any knot feature is introduced.

12.5 Corrected hypothesis and non-fitting rule

The next candidate question is whether a frozen topological symmetry class has a relation to the intrinsic shell involution, particularly to self-dual half filling. A valid test must be specified before knot labels and shell labels are jointly inspected. At minimum it must satisfy:

1. shell features are generated from (l, k) and independently calculated control observables;
2. knot features are generated from a frozen canonical knot table;
3. the comparison rule is invariant under mirror convention where intended;
4. particle–hole partners are fixed by $k \mapsto C_l - k$, never by post-hoc element pairing;
5. permutation and replacement-anchor null controls are run before interpretation;
6. failure is retained without changing $z_\star$ or the selected knot family to improve a score.

12.6 Relation to the N-body shell formalism

Chapter 11 defines the dynamical shell record Σ_{nlk} with explicit electron–nucleus and electron–electron coordinates. The knot diagnostic does not replace that record. Instead, a future map would have to be a derived functional

$$\mathcal{K} : \Sigma_{nlk}[\gamma] \longrightarrow \mathcal{C}_{\text{topo}} \quad (12.8)$$

whose domain, projection dependence and invariance properties are stated before physical interpretation. In unrestricted three-dimensional electron dynamics, exchange topology remains permutation-based; braid invariants require an effective two-dimensional or otherwise constrained sector.

12.7 Current verdict

The conformal knot machinery is implemented as an exploratory diagnostic and has produced one useful negative control and one better-posed shell-level hypothesis. No knot-shell physical law is validated. The next gate is a shell-only particle-hole test across principal quantum number, followed by a pre-registered symmetry-class comparison against the frozen knot descriptors.

Chapter 13

From Nuclear Centres to Molecular Shell Dynamics

13.1 Why the nucleus must remain in the model

A shell topology detached from its nuclei would discard the dominant attractive structure that organizes ordinary atomic and molecular bound states. The multicentre extension therefore starts from the standard clamped-nuclei electronic Hamiltonian

$$H_e = \sum_{i=1}^{N_e} \left[-\frac{1}{2} \nabla_i^2 - \sum_{A=1}^M \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} \right] + \sum_{i < j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + V_{NN}, \quad (13.1)$$

with

$$V_{NN} = \sum_{A < B} \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|}. \quad (13.2)$$

The nuclei $\mathcal{A}_A = (Z_A, \mathbf{R}_A)$ are attractive Coulomb centres for the electrons. In the exact conservative Hamiltonian they are not dissipative attractors; in an iterative electronic-structure algorithm, however, a converged fixed point may possess a basin of attraction in the numerical state space. The two meanings are kept distinct.

13.2 Shells as local organization, bonding as global reorganization

An isolated-atom configuration provides a useful local shell decomposition, but a molecular eigenstate is not generally a tensor product of immutable atomic shells. Bond formation changes the global one-particle density matrix, orbital occupations, exchange structure and correlation. The shell-level programme therefore uses atomic subshells as incoming coordinates and asks how their control observables reorganize when the nuclear-centre set changes from

$$\{\mathcal{A}_A\}_{\text{separated}} \longrightarrow \{\mathcal{A}_A\}_{\text{molecule}}. \quad (13.3)$$

A bond is not defined here as a knot. The control definition remains energetic and electronic: a molecular state is characterized by its Hamiltonian, density, reduced density matrices and response to nuclear geometry. Topological descriptors may be appended only as derived diagnostics.

13.3 Multicentre winding diagnostics

For a declared oriented projection $P : \mathbb{R}^3 \rightarrow \mathbb{C}$, an electron path may be measured relative to each nuclear centre,

$$w_{iA}[\gamma] = \frac{1}{2\pi} \oint_{\gamma} d \arg P(\mathbf{r}_i - \mathbf{R}_A), \quad (13.4)$$

and relative electron paths may be measured by

$$w_{ij}[\gamma] = \frac{1}{2\pi} \oint_{\gamma} d \arg P(\mathbf{r}_i - \mathbf{r}_j). \quad (13.5)$$

For $M > 1$ nuclei, the vector $\mathbf{w}_i = (w_{i1}, \dots, w_{iM})$ can distinguish projected circulation around different centres. These quantities are projection-dependent unless an effective two-dimensional physical constraint is separately justified.

13.4 Basin transfer as a reduced diagnostic

In a reduced orbital or density-flow description, one may define centre-weighted populations using a fixed partition of real space, for example

$$Q_A = \int_{\Omega_A} \rho(\mathbf{r}) d^3r, \quad (13.6)$$

with the partitioning rule specified independently of the desired chemical conclusion. A bond-formation path then induces a redistribution

$$\Delta \mathbf{Q} = (\Delta Q_1, \dots, \Delta Q_M). \quad (13.7)$$

A future topological descriptor may be compared with this redistribution, but it cannot replace charge conservation or create a bond where the control wavefunction does not support one.

13.5 Adiabatic and nonadiabatic paths

For a nuclear path $\mathbf{R}_A(\lambda)$, an adiabatic electronic branch $|\Psi(\lambda)\rangle$ may acquire a geometric phase

$$\Gamma = i \oint \langle \Psi(\lambda) | \partial_{\lambda} \Psi(\lambda) \rangle d\lambda. \quad (13.8)$$

This is conceptually distinct from projected particle winding. Any combined shell-topology record must therefore keep Berry/Pancharatnam holonomy, exchange parity and projected spatial winding as separate fields rather than collapse them into one scalar.

13.6 Candidate molecular state record

A minimal multicentre extension of Chapter 11 is

$$\Sigma_{\text{mol}} = (\{Z_A, \mathbf{R}_A\}, \rho, \gamma, \{n_k\}, \pi_{\text{exch}}, \{w_{iA}\}, \{w_{ij}\}, \Gamma, \mathcal{P}_{\text{shell}}), \quad (13.9)$$

where $\mathcal{P}_{\text{shell}}$ records the chosen shell/subshell projection and its provenance. The density ρ , one-body density matrix γ and natural occupations n_k are control observables; the winding and conformal fields are candidate diagnostics.

13.7 From one structural candidate to a competing-state ensemble

The compound trajectory exposed a structural problem that a single molecular graph cannot represent safely. The same stoichiometry may support qualitatively different electronic or intermolecular realizations. In particular, the closed-shell XY_2 study contains an activated linear three-centre/four-electron candidate and weak $X \cdots Y_2$ complexes under the same empirical formula.

Version v0.13 therefore replaces the mapping

$$\text{composition} \longrightarrow \text{one predicted graph} \quad (13.10)$$

with

$$\text{composition} \longrightarrow \text{unranked state ensemble} \longrightarrow \text{physical admission}. \quad (13.11)$$

For each of the nine frozen $\text{Ne/Ar/Kr} \times \text{F/Cl/Br}$ compositions, the current ensemble contains

1. ACTIVATED_LINEAR_3C4E,
2. WEAK_COMPLEX_LINEAR_END_ON,
3. WEAK_COMPLEX_T_SHAPED.

This produces 27 state candidates. Their prior rank and prior probability are explicitly absent. Enumeration is not a stability prediction, and reference knowledge about KrF_2 or rare-gas-halogen van-der-Waals complexes is not converted into a prior score.

13.8 First common molecular-state relaxation control: v0.14A/A1

The first executable consumer of the v0.13 ensemble is a conventional low-cost relaxation screen. It is not a new Resonant-Chemistry Hamiltonian. The frozen electronic-structure policy uses B97M-V with VV10 nonlocal correlation, the def2-TZVPD basis, neutral singlets, PySCF and geomeTRIC, and one common SCF/optimizer policy.

The original preregistration used a common numerical-grid policy. A backend smoke then exposed a PySCF SG-1 pruning limitation while constructing the KrF_2 nonlocal-correlation grid. No KrF_2 energy had yet been produced. The v0.14A1 amendment therefore changed only the pruning algorithm, globally for both regular and nonlocal grids, to `nwchem_prune`. The functional, basis, grid sizes, charge/spin policy, optimizer, SCF thresholds and seed geometries remained frozen. This amendment is retained separately rather than rewritten into the original preregistration.

The amended backend smoke passed, and F_2 , Cl_2 and Br_2 were then relaxed from the same 2.0 Å starting separation. Their same-method equilibrium distances are used only as numerical seed scales. Each XY_2 composition receives five frozen starts: three symmetric activated-linear starts with $r_{XY}/r_{YY} = 1.0, 1.3, 1.6$, one weak end-on start, and one weak T-shaped start. The complete preregistered matrix therefore contains 45 XY_2 relaxations.

13.9 Partial v0.14A1 execution evidence

The first full workflow was cancelled after the common backend and ligand-dimer stages and after eight of the nine formula jobs had produced receipts. Those first-attempt formula receipts were recovered and persisted unchanged. The durable checkpoint is therefore

$$8/9 \text{ formulae}, \quad 40/45 \text{ frozen starts}. \quad (13.12)$$

The missing composition is ArBr_2 . Its status is `MISSING_EXECUTION`; no chemical negative label is inferred from cancellation or missing runtime evidence.

For all eight completed compositions, the lowest successful *screening* electronic energy belongs to one of the weak-complex seed families. Because no Hessian has been evaluated, this does not establish a ground state or even harmonic local-minimum status.

The activated-start survival pattern itself is retained as a threshold-free diagnostic. Representative observations are:

- KrF_2 : three of three activated starts complete and converge to the same stored-precision symmetric basin, with $r_{\text{Kr-F}} \approx 1.883 \text{ \AA}$; this basin lies about 9.288 kcal/mol above the lowest successful weak-linear screening state under the same method;
- ArF_2 : three of three activated starts complete, with an activated-to-lowest-weak screening gap of about 37.99 kcal/mol;
- NeF_2 : two of three activated starts complete, with a corresponding gap of about 95.93 kcal/mol;
- NeCl_2 and NeBr_2 : none of the three activated starts complete, while both weak seed families complete;
- ArCl_2 , KrCl_2 and KrBr_2 retain respectively two, two and one successful activated starts out of three.

These values are descriptive screen outputs only. No activation law, threshold, topology classifier or seed-specific rescue rule is fitted from them.

A later rerun request re-executed a broader GitHub matrix scope than intended. To preserve provenance, the original first-attempt receipts remain the v0.14A1 evidence for the eight already-completed formulae. Only a future ArBr_2 receipt may fill the missing slot without replacing those first-attempt records.

13.10 Electronic topology is a separate admission problem

A linear symmetric geometry is not by itself evidence for a three-centre/four-electron electronic topology, just as an XY_2 formula is not such evidence. The v0.10 admission contract therefore keeps three independent diagnostic families separate: orbital-subspace evidence, real-space/force evidence, and fragmentation/isomer-energy evidence. A topology label requires independent agreement under that contract; insufficient evidence remains UNKNOWN.

The v0.14A/A1 relaxation screen supplies candidate geometries and energies to later controls, but it does not replace v0.10.

13.11 Simple-system molecular validation ladder

Simple systems remain necessary controls even though they are no longer the first molecular code implemented in the repository:

1. H_2^+ as a one-electron two-centre control;
2. H_2 as the first exchange/correlation bond;
3. HeH^+ as an asymmetric two-centre test;
4. LiH as a core–valence separation test;
5. only then more complicated multi-centre or ring systems where loop descriptors are especially vulnerable to confirmation bias.

This control ladder is complementary to the v0.14 closed-shell XY_2 screening trajectory rather than its historical predecessor in implementation order.

13.12 Current status and next gate

The repository now contains an implemented conventional molecular-state relaxation layer and partial v0.14A1 execution evidence, but it still does *not* contain a validated Resonant-Chemistry molecular topology or molecular-energy law.

The immediate gate is operationally narrow: complete the missing ArBr_2 execution under the same frozen v0.14A1 method and seed policy. Only after all nine formula receipts and all 45 frozen starts are durably present may a separate v0.14B Hessian/local-minimum protocol be frozen. A timeout, cancellation, SCF failure or optimizer failure remains explicit evidence and does not license target-specific rescue.

After local-minimum admission, conventional energetic comparison and the independent v0.10 electronic-topology diagnostics may be applied. TIR, PhaseNav and affective semantics remain outside the control electronic energies unless a separately defined no-refit observable passes its own admission gate.

Part VI

Relational Admission Layer

Chapter 14

Admission Contract for TIR, PhaseNav, and Resonant Semantics

14.1 Why a separate admission layer is necessary

The control chapters deliberately keep TIR, PhaseNav and affective semantics out of the atomic and molecular control Hamiltonians. This is not a rejection of the relational programme; it is the condition that makes a later test identifiable. If a candidate correction is introduced before the conventional control is frozen, agreement with experiment cannot distinguish the new mechanism from ordinary orbital relaxation, exchange, correlation, basis/grid effects or numerical tuning.

The admission rule is therefore

$$\text{control PASS} \wedge \text{observable defined} \wedge \text{no-refit prediction} \wedge \text{null controls} \implies \text{candidate relational test.} \quad (14.1)$$

It does not imply physical promotion.

The same discipline applies when the conventional control itself is partial. A missing runtime result, such as the current ArBr_2 slot in v0.14A1, remains UNKNOWN/MISSING EXECUTION and is not filled by a relational prediction.

14.2 TIR metadata and the information quantum

The project carries

$$\kappa = \frac{\ln 2}{24\pi} \quad (14.2)$$

as a TIR-defined quantity. In the atomic and molecular control calculations presented in this book, κ is metadata: it does not alter hydrogenic, Hartree–Fock, CI, DFT, spin–orbit, geometry-optimization or shell-topology control energies. Any future energy or transition formula containing κ must state the dimensional map that makes the term admissible and must be tested against a control that did not use the target value in its construction.

14.3 Relation operators

For active electronic subspaces $\mathcal{E}_A$ and $\mathcal{E}_B$, a general relational operator may be written

$$W_{AB} : \mathcal{E}_B \rightarrow \mathcal{E}_A. \quad (14.3)$$

This notation alone does not determine a physical interaction. An admissible W_{AB} must provide: a domain and codomain, covariance or transformation rule, dimensional units where applicable, a construction from control observables, and a falsification criterion. A semantic label or learned similarity score is not sufficient to promote W_{AB} into the physical Hamiltonian.

14.4 PhaseNav/36D as representation and orchestration interface

A PhaseNav vector may be useful as a deterministic representation of a structured record. For a shell state we write abstractly

$$\mathcal{V}_{36} : \Sigma_{nlk} \longrightarrow \mathbb{T}^{36} \quad (14.4)$$

only after the encoder is frozen. The dimension 36 is treated here as a project representation width. A 36-dimensional encoding does not by itself prove that physical electron dynamics has 36 spatial or internal dimensions. Distances, winding or coherence measures in the encoded space must be distinguished from invariants of the underlying quantum state.

A valid encoder must therefore satisfy at least deterministic replay, explicit feature provenance, stable treatment of missing values, and tests showing which transformations of the physical state should or should not leave the representation invariant.

PhaseNav may also provide the native command, gate and receipt surface for project execution. The intended layering is

$$\text{command} \rightarrow \text{concept} \rightarrow \text{gate} \rightarrow \text{36D state} \rightarrow \text{backend adapter} \rightarrow \text{receipt}. \quad (14.5)$$

This is an operational architecture, not permission for the representation layer to alter a conventional result. If PySCF, a Hartree–Fock solver, a Hessian calculation or another control backend returns an energy, convergence failure or exception, a PhaseNav command must preserve that outcome and its provenance. Backend-specific code is therefore isolated behind adapters rather than allowed to define the relational semantics.

14.5 Repository authority, NOEMA and Library mirrors

For Resonant Chemistry, the GitHub repository default branch `main` is the project Source of Truth for canonical code, manuscript, benchmark and gate state. A research branch may contain newer candidate work, but it is not canonical merely because it exists.

NOEMA/Library may store mirrors, recovery copies, semantic cards, receipts, validation states, pointers and project continuity information. Those roles are operational and provenance-oriented. They must not become a runtime dependency that prevents checkout, build, test or recovery of the canonical repository when Library access is absent. The normal persistence direction for canonical repository state is therefore

$$\text{repo canon} \longrightarrow \text{Library/NOEMA mirror and index}. \quad (14.6)$$

A Library-to-repo recovery is exceptional: it must verify artifact identity/provenance before reintroducing missing material and must not silently overwrite newer repository state.

A NOEMA or Library receipt can establish what calculation, data and code state were used; it cannot by itself establish an external physical effect. The scientific manuscript must remain reproducible from equations, repository code and frozen benchmark inputs independently of the conversational runtime that helped assemble it.

14.6 Affective semantics

Affective mappings are explicitly outside the atomic and molecular control Hamiltonians. If a future map

$$\mathcal{F}_{\text{aff}} : \mathcal{I}(\Psi) \rightarrow \mathcal{A} \quad (14.7)$$

consumes state invariants, its output is a semantic/operational coordinate unless separate evidence supports a physical interpretation. The map may consume immutable occupation spectra, entropies, multiplet descriptors or relational observables; it may not rewrite them. This preserves the distinction between measured/computed state and interpretation.

14.7 Required prospective test pattern

A candidate relational observable R should be developed in the following order:

1. select a physical system and freeze its conventional control protocol;
2. construct R from inputs unavailable to no other tuned parameter;
3. freeze the formula and all constants before opening the target reference where possible;
4. predict a new quantity, ordering, ratio or failure boundary;
5. run ablations and replacement-anchor/null controls;
6. retain failure without changing the formula after the target is known.

A successful retrospective reconstruction can motivate a candidate, but it is not equivalent to a prospective test.

14.8 Compound-state admission example

The v0.13/v0.14 trajectory illustrates the separation in executable form. v0.13 enumerates competing activated and weak-complex states without ranking them. v0.14A/A1 supplies a common conventional DFT relaxation screen. The screen may reveal distinct basins or failed starts, but it does not assign electronic topology. A later Hessian gate must establish local-minimum status, and the v0.10 electronic-topology contract still requires independent diagnostic families before a 3c4e/VDW label is admitted.

Thus a relation-state name is a candidate coordinate, not a physical verdict. Physical controls may select one state, several metastable states, or reject all currently enumerated states.

14.9 Current admission status

The atom, shell, compound and partial molecular-control layers through this book are sufficiently explicit to host candidate relational observables, but no TIR, PhaseNav or affective correction is promoted into their Hamiltonians or molecular energies. The conformal shell programme remains a candidate diagnostic. The compound-state ensemble is implemented but unranked. The v0.14A1 molecular screen is partial at eight of nine formulae and has no Hessian/topology admission.

The next admissible relational-physics step remains a no-refit observable whose prediction differs from the conventional control in a pre-declared way. Operational improvements to PhaseNav commands, receipts or 36D encoding are valuable software work, but they are not substitutes for that scientific admission test.

Part VII

Reproducibility and Project State

Appendix A

Benchmark and Validation Ledger

This appendix summarizes the machine-readable checkpoints that support the narrative chapters. The JSON files in `benchmarks/` remain the authoritative numerical records; the table below is an index, not a replacement for them. A benchmark name identifies a local gate and does not promote a broader physical interpretation.

Artifact	Status class	Scientific role
BLIND_MASS_SOLVE_S_SHELL_VO_1.json	tested control	Blind/reference-isolated <i>s</i> -shell mass-solving checkpoint for the first one- to four-electron systems.
BLIND_MULTI_L_H_TO_NE_VO_1.json	tested control	Common multi- <i>l</i> Hartree–Fock checkpoint from H through Ne; ten of ten systems converged.
NA_TO_AR_DIIS_VO_1.json	tested control	Global DIIS stabilization for the third period without element-specific tuning.
H_TO_KR_CONTROL_MANIFEST_VO_1.json	coverage control	Provenance and acceptance manifest for neutral H through Kr.
K_TO_KR_ROBUST_QUALITY_VO_1.json	tested control	Adaptive quality ladder for the K–Kr region with virial rejection of false stationary points.
ANGULAR_MULTIPLETT_P_D_VO_1.json	angular control	Equivalent-shell <i>LS</i> multiplet structure for <i>p</i> and diagnostic <i>d</i> shells.
SPIN_ORBIT_J_SEQUENCE_VO_1.json	angular control	Weak-coupling <i>J</i> ordering derived from the determinant-space spin–orbit operator.
PERIOD2_ATOM_SPECIFIC_SPECTROSCOPY_VO_1.json	approximate spectroscopy control	Radial F^k , Pauli central-field ζ , and B–Ne level structure; post-hoc references are isolated from the solve.
PERIOD2_ACTIVE_P_CI_VO_1.json	correlation control	Finite active- <i>p</i> CI for period-2 term centers.
PERIOD2_CORRELATED_SPECTRUM_VO_1.json	correlation control	Radiagonalized spin–orbit structure in the correlated active space.
CARBON_ACTIVE_SPACE_CONVERGENCE_VO_1.json	diagnostic	Radial- <i>p</i> active-space convergence for carbon.
CARBON_CORRELATION_CHANNEL_DIAGNOSTIC_VO_1.json	diagnostic	Separation of radial- <i>p</i> , explicit- <i>s</i> , and other channel effects on 1D and 1S .
CARBON_PD_CHANNEL_DIAGNOSTIC_VO_1.json	diagnostic	Virtual- <i>d</i> channel isolation.
CARBON_SPARSE_MULTICHANNEL_DIAGNOSTIC_VO_1.json	diagnostic	Sparse multichannel CI with thousands of determinants; establishes the frozen-orbital boundary.

Artifact	Status class	Scientific role
SHELL_NBODY_TOPOLOGY_P_SHELL_VO_1.json	candidate implementation	Particle-hole duality and shell-level topology primitives; physical trajectory validation remains open.
KNOT_CONFORMAL_ATOM_SYMMETRY_VO_1.json	historical candidate	Original exploratory knot/atom comparison retained for provenance; direct spectral matching was not promoted and the comparison object is superseded by shell-level v0.2.
COMPOUND_STATE_ENSEMBLE_VO_13.json	candidate architecture	Nine closed-shell XY_2 compositions expanded into 27 unranked activated/weak-complex state candidates; enumeration is not stability or topology admission.
MOLECULAR_STATE_RELAXATION_PREREG_VO_14A.json	preregistered control	Frozen common B97M-V/VV10/def2-TZVPD relaxation screen, five starts per XY_2 composition and no Hessian/topology verdict.
MOLECULAR_STATE_RELAXATION_AMENDMENT_VO_14A1.json	numerical amendment	Post-smoke/pre-screening global replacement of SG-1 pruning by common NWChem pruning on regular and NLC grids; no element-specific rescue.
MOLECULAR_BACKEND_SMOKE_VO_14A1_EXECUTION.json	execution receipt	Passing amended F_2 + KrF_2 energy/gradient backend compatibility smoke; software/method compatibility only.
MOLECULAR_DIMER_PREPASS_VO_14A1_EXECUTION.json	execution receipt	Same-method $F_2/Cl_2/Br_2$ relaxation prepass used only to generate internal seed scales.
MOLECULAR_STATE_RELAXATION_PARTIAL_READOUT_VO_14A1.json	partial execution evidence	Durable first-attempt aggregate for eight of nine formulae and 40 of 45 frozen starts; $ArBr_2$ remains missing execution.
MOLECULAR_STATE_RELAXATION_ACTIVATED_SURVIVAL_MATRIX_VO_14A1_PARTIAL.json	descriptive diagnostic	Threshold-free activated-start survival and activated-to-weak screening gaps; explicitly no fitted law or topology classifier.
MOLECULAR_STATE_RELAXATION_RERUN_SCOPE_VO_14A1.json	provenance guard	Records that a rerun executed broader matrix scope than intended and preserves original first-attempt receipts as the checkpoint evidence.

A.1 Interpretation rule

A benchmark status is local to the declared model and gate. H–Kr coverage does not imply high-precision term-resolved spectroscopy for every element, a passing shell-topology unit test does not validate a topological law of chemistry, and a successful molecular relaxation does not establish a harmonic minimum or electronic topology. The most restrictive scientifically relevant caveat accompanies the result throughout the book.

A missing execution receipt is not a failed chemical hypothesis. Conversely, a complete software workflow does not promote a scientific candidate. Runtime cancellation, SCF failure, optimizer failure, negative controls and falsifiers are retained as distinct status classes.

A.2 Current correlation frontier

The first one-coordinate state-averaged active-external carbon gate has produced a strict decrease in its frozen equal-weight state-average objective and is therefore variationally informative along the tested direction. The next correlation control is multidimensional state-averaged orbital optimization/MCSCF-type relaxation without experimental term-energy fitting, followed by correlated spin-orbit re-evaluation only after that orbital layer is frozen.

A.3 Current molecular frontier

The molecular state-ensemble architecture is implemented and the v0.14A1 common relaxation screen has durable partial execution evidence for eight of nine formulae. The immediate gate is not a Hessian fit or a new topology rule: it is completion of the missing ArBr_2 execution under the same frozen method and seed policy. Only after a complete 9/9, 45/45 relaxation ledger may a separately preregistered Hessian/local-minimum gate be opened.

Appendix B

Equation-to-Code Map

The purpose of this appendix is to make the textbook executable in the scientific sense: a reader should be able to locate the code that realizes each major model class and the tests or receipts that constrain it. Paths refer to the repository root.

Model layer	Implementation	Primary verification
Atomic identity and configurations	<code>reschem/atom.py</code> and associated configuration helpers	hydrogenic/atom constructor tests and H–Kr manifest
Hydrogenic radial control	radial Schrödinger/atom foundation modules	analytic $E_n \propto -Z^2/n^2$ comparison
Helium variational	<code>reschem/helium.py</code>	variational-energy and virial tests
Helium RHF	<code>reschem/rhf_helium.py</code>	RHF regression and post-hoc HF-limit comparison
Helium CI	<code>reschem/ci_helium.py</code>	finite-CI energy lowering and state-invariant tests
Semantic atom cards	<code>reschem/semantic_cards.py</code> , <code>semantic_cards/ATOM_CARD_INDEX_CURRENT.json</code>	H–Kr indexed coverage; interpretive fields remain unassigned without provenance
Lithium open shell	<code>reschem/uhf_lithium.py</code>	energy, S^2 and spin-contamination tests
Blind s -shell batch	<code>reschem/mass_solver_s_shell.py</code>	blind batch benchmark and convergence tests
Multi- l atomic HF	<code>reschem/atomic_hf_average.py</code>	H–Ne reference-isolated checkpoint
Global SCF stabilization	<code>reschem/atomic_hf_diis.py</code>	Na–Ar and K–Kr quality gates
H–Kr batch interface	<code>reschem/atomic_mass_h_to_kr.py</code>	coverage manifest and virial/convergence ledger
Equivalent-shell multiplets	angular multiplet solver modules	p^n/d^n microstate dimensions and LS term tests
Spin-orbit J ordering	spin-orbit angular solver modules	frozen J sequences for p and d shells
Period-2 spectroscopy	atom-specific radial spectroscopy modules	F^k , ζ and B–Ne spectrum benchmark
Period-2 active CI	<code>reschem/period2_active_ci.py</code>	active-space CI tests and benchmark
Correlated spectrum	<code>reschem/period2_correlated_spectrum.py</code>	correlated spin-orbit regression
Carbon channel diagnostics	<code>reschem/carbon_*ci.py</code> family	convergence/channel/sparse benchmark suite
Carbon state-average p relaxation	<code>reschem/carbon_state_averaged_orbital_relaxation.py</code>	active-external orthonormality, fixed equal-weight $^3P/^1D/^1S$ objective, and variational scan gate

Model layer	Implementation	Primary verification
Knot conformal exploration	<code>reschem/knot_conformal_analysis.py</code>	conformal utility tests; direct spectral interpretation rejected by null control
Shell N-body topology	<code>reschem/shell_nbody_topology.py</code>	shell-capacity, duality, winding, parity and conformal-inverse tests
Shell symmetry v0.2	<code>reschem/shell_symmetry.py</code>	particle-hole involution tests independent of atom labels
Compound relation skeleton	<code>reschem/compound_shell_relations.py</code> , <code>reschem/relation_graph.py</code>	v0.1/v0.3 structural candidate tests and benchmark ledgers
Three-centre relation mode	<code>reschem/hyperrelation.py</code>	B ₂ H ₆ /bridged-halide structural bookkeeping tests; no literal fractional-electron claim
Coordination/reorganization	<code>reschem/coordination_ladder.py</code> , <code>reschem/parity_validation.py</code>	parity-matched v0.6 null and ladder regression tests
Energetic admission contract	<code>reschem/energetic_admission.py</code>	fail-closed data structures and preregistered v0.8 reaction/method contract
Closed-shell activation	<code>reschem/closed_shell_activation.py</code>	v0.9 nine-composition atlas and no-element-special-case tests
Electronic topology admission	<code>reschem/electronic_topology.py</code>	independent diagnostic-family aggregation tests; insufficient evidence remains UNKNOWN
Competing compound-state ensemble	<code>reschem/compound_state_ensemble.py</code>	<code>benchmarks/COMPOUND_STATE_ENSEMBLE_V0_13.json</code> ; 27 unranked states and no prior scores
Semantic entity projection	<code>reschem/semantic_projection.py</code> , <code>reschem/entity_registry.py</code>	deterministic 231 compound cards + 27 emergent state cards; relation, lineage and physical-holonomy boundary tests
Current repository card loader	<code>reschem/repository_cards.py</code>	indexed 36 atom bases, nondestructive overlays, model/state entities and hashed calculation contexts
Common molecular-state relaxation	<code>reschem/molecular_state_relaxation.py</code>	v0.14A preregistration, v0.14A1 backend/dimer receipts, and partial eight-of-nine formula readout
Molecular semantic projection	<code>reschem/molecular_semantic_projection.py</code>	model + nine formula cards must agree exactly with machine-readable v0.14A1 execution status
Semantic-card coverage gate	<code>scripts/audit_semantic_card_coverage.py</code>	H–Kr coverage, model/gate coverage, generated entity counts, relation/holonomy provenance and no silent semantic promotion
Molecular execution provenance wrapper	<code>scripts/wrap_molecular_relaxation_v0_14a1.py</code>	<code>tests/test_molecular_execution_wrapper.py</code> and rerun-scope provenance guard

B.1 Control backend boundary

The v0.14 molecular relaxation module uses PySCF/geomeTRIC as conventional numerical backends. Those packages do not define the Resonant-Chemistry relational state taxonomy, gate semantics or scientific promotion rules. The project layer defines the frozen method policy, common seed generation, failure preservation and receipts; the backend supplies conventional electronic energies, gradients and geometry optimization.

Future orchestration may expose the same operations through a PhaseNav-native command/-gate surface. Such orchestration must preserve the conventional results and provenance and may not rewrite a backend failure or energy to satisfy a relational hypothesis.

B.2 Semantic calculation boundary

The semantic entity registry is an input-organizing layer for calculations. It records properties, state invariants, source artifacts, graph relations and provenance lineage. A hashed card context can therefore be attached to a numerical run so that the semantic/entity state consumed by the run is reproducible.

Registry membership does not change the scientific status of a card. In particular, a model-derived emergent entity remains a candidate until an independent physical admission gate is satisfied. Provenance holonomy is a lineage record; physical holonomy requires separately computed winding/phase/topology observables with provenance.

B.3 No hidden physical authority

A function name, semantic-card field or 36D encoding is not granted physical authority by appearing in this map. The equation in the textbook and the benchmark acceptance rule jointly define what the implementation is supposed to compute. If code and documentation diverge, the discrepancy is a defect to be resolved rather than permission to reinterpret the equation silently.

B.4 Reproducibility convention

Every new solver or candidate module should ship with: a deterministic or tolerance-bounded test; a machine-readable benchmark/receipt when numerical output is scientifically discussed; a semantic-card projection for each newly addressable entity/model/state; a monograph section giving the governing equations and approximations; and provenance sufficient to identify the tested implementation. Exploratory notebooks or dashboards may visualize results, but they are not the sole repository of a scientific claim.

Merged changes that alter implementation status, validation state, semantic-card coverage or the current open gate must trigger documentation reconciliation. ‘main’ is the project Source of Truth; historical research branch names are not a substitute for current repository state.

Appendix C

Validation, Reproducibility, and Repository Contract

C.1 Validation hierarchy

Software correctness and scientific validity are recorded separately. A useful minimum hierarchy is

syntax/build $\rightarrow$ unit test $\rightarrow$ numerical regression $\rightarrow$ control benchmark $\rightarrow$ external comparison $\rightarrow$ prospective
(C.1)

A PASS at one level does not imply a PASS at the next. For example, the shell-topology unit tests verify algebraic implementation primitives but not a physical knot-shell relation, and a converged molecular geometry optimization is not by itself a harmonic-minimum or electronic-topology validation.

C.2 Reference isolation

Whenever the target quantity can influence model design, the repository records whether the reference was available to the solver. A blind or reference-isolated calculation freezes its Hamiltonian, basis policy and convergence gate before the external value is opened. If reference data were already known to the researchers, the implementation may still be reference-isolated, but the text does not call the human process blind.

The same rule applies to molecular state enumeration. Already-opened knowledge about KrF_2 or rare-gas-halogen van-der-Waals complexes may motivate a falsifiable architecture, but it is not converted retrospectively into a prior rank for v0.13 states.

C.3 Negative, partial, and missing results

A negative result is part of the state of the theory. The direct knot-gap/atom-gap comparison failed a random-partition null test and remains recorded. Carbon active-space growth shows that a larger fixed CI can improve one term and worsen another. The robust atomic solver shows that a numerically larger fallback stage can degrade virial quality for a system that already passed an earlier stage. The v0.11/v0.12 atomic finite-difference route is retained even though it is not admitted as a molecular chemistry classifier.

Partial execution is a separate status class. In v0.14A1, eight of nine XY_2 formula jobs produced durable first-attempt receipts, yielding 40 of 45 frozen starts. The missing ArBr_2 job is

labelled `MISSING_EXECUTION`. A missing or cancelled calculation is not converted into a negative chemical result.

Likewise, a returned stationary geometry in the relaxation screen is not labelled a local minimum before an independently frozen Hessian gate. The lowest successful electronic energy among the sampled starts is called a screening result, not a validated ground state.

C.4 First-attempt and rerun provenance

A rerun is not allowed silently to replace frozen first-attempt evidence. The v0.14A1 recovery illustrates this rule: a GitHub matrix rerun executed a broader scope than the missing ArBr_2 job alone. The project therefore retains the original first-attempt receipts for the eight already-completed formulae and records the rerun scope separately. A future ArBr_2 result may fill the missing slot, but does not rewrite the earlier eight receipts in that checkpoint.

This convention separates reproducibility from cherry-picking. Repetition may test stability, but the first declared result remains identifiable.

C.5 Preregistration amendments

A frozen protocol may require a technical amendment before scientific outputs exist. Such a change is not rewritten into history. The v0.14A backend smoke exposed an SG-1 grid-pruning implementation boundary for krypton before a KrF_2 energy was produced. The v0.14A1 amendment changed the pruning algorithm globally on both regular and nonlocal grids, retained all other method and seed settings, and was recorded as *post-smoke / pre-screening-output*. The original preregistration and the amendment therefore remain separately auditable.

Target-specific rescue is distinct and remains forbidden. A change made only because one composition gives an inconvenient scientific result would require a new declared experiment rather than a silent repair of the frozen gate.

C.6 Semantic entity-card reproducibility

Semantic cards are part of the calculation state. A newly addressable atom, compound candidate, relational state, molecular screen or later emergent entity must have a card or deterministic projection that records its properties, state invariants, source artifacts, relation edges and epistemic status.

The current repository distinguishes two notions of holonomy. *Provenance holonomy* is deterministic lineage: parent-card identifiers, generating operation and a lineage hash. *Physical holonomy* is reserved for explicitly computed winding, phase, connection or topology observables. A semantic graph loop or provenance loop is not evidence of physical holonomy.

High-cardinality entities may be generated deterministically from frozen model generators instead of being copied manually into hundreds of files. The generator itself is tested against frozen cardinalities and the current benchmark. Stable/base cards and model/gate checkpoints remain persisted under `semantic_cards/`.

A calculation can request a semantic context from the current repository card registry. The selected cards and, optionally, their relation targets are canonically serialized and hashed. The hash identifies the consumed semantic/entity context; it does not prove that two contexts are physically equivalent.

Emergent model outputs remain explicitly candidate. The registry status `MODEL_DEFINED_EMERGENT_CANDIDATE`

means only that a deterministic model generated an addressable entity. It does not establish molecular existence, harmonic stability, ground-state status, electronic topology or project canon.

C.7 Semantic-card anti-drift gate

The maintained compound/molecular CI runs a card-coverage audit. The gate checks:

1. canonical H–Kr neutral atomic-card coverage;
2. model/gate card coverage through the current compound/molecular frontier;
3. deterministic generation of the 231 v0.1 compound candidates and 27 v0.13 relational states;
4. exact agreement between the nine v0.14A1 formula cards and the machine-readable expected/completed formula sets;
5. preservation of ArBr₂ as missing execution rather than chemical failure;
6. absence of silent TIR or affective assignments without provenance;
7. provenance for any non-empty physical-holonomy record;
8. no semantic promotion of Hessian status, ground-state ranking or electronic topology beyond the underlying evidence.

Thus a new scientific/model entity cannot be merged merely by updating code while leaving the calculation-card layer stale.

C.8 LaTeX as the formal scientific surface

The scientific architecture is maintained as this single LaTeX textbook. Every major code path, benchmark and candidate formalism must be represented here at its actual epistemic status. Supplemental Markdown can contain operational detail, but a scientific definition or validation rule that affects interpretation may not exist only in a chat transcript or dashboard.

A successful PDF build proves that the LaTeX source is buildable; it does not prove that the source is current relative to the repository. Therefore documentation reconciliation is a separate requirement: when a merged change alters implementation status, benchmark evidence, semantic-card coverage, nonclaims or the next open gate, `README.md`, `STATUS.md`, the relevant textbook chapter and its appendices must be reviewed against the new `main` state.

C.9 Repository Source of Truth

The GitHub repository default branch `main` is the project Source of Truth. Research branches may contain candidate work and reconciliation patches, but a branch name from an earlier workstream is not current canon merely because an older workflow still references it.

Continuous-integration triggers for the maintained textbook, bibliography and compound/molecular regression surfaces must therefore validate `main` and pull requests targeting `main`. Manual dispatch may be retained for reproducibility, but historical branch-only triggers are not an adequate release contract.

C.10 Continuous-integration build

The maintained textbook gate runs Python test discovery, repository-structure checks, the live bibliography audit, LaTeX/BibTeX compilation, unresolved-reference rejection, hashes and PDF artifact upload. The dedicated bibliography gate independently compiles the modular bibliography

when manuscript/bibliography files change. The compound/molecular regression gate parses the relevant JSON ledgers, verifies current v0.13/v0.14 surfaces and enforces semantic-card coverage/epistemic separation.

A CI infrastructure or account/billing failure is reported separately from software/scientific failure. Absence of a workflow run is not a PASS.

C.11 Directory completeness

Git does not represent genuinely empty directories. Therefore the repository cannot prove the existence of an intended-but-empty folder merely from a tree listing. The project policy is stronger: every required directory should contain a real artifact or a descriptive `README.md` specifying purpose, owner/status and the gate that will populate it. Silent placeholder-only directories are not considered complete.

The expected top-level content areas are populated: `THEORY/`, `benchmarks/`, `docs/`, `monograph/`, `reschem/`, `tests/`, `semantic_cards/`, `web/`, and `.github/workflows/`. The modular bibliography under `monograph/bibliography/` is also part of the required manuscript surface.

C.12 Branch and promotion discipline

Research increments are developed on dedicated branches. A branch may contain implemented and tested candidates without changing project canon. Merge to the default branch and promotion of a physical interpretation are separate decisions. A commit hash proves the identity of repository content; it does not by itself promote the scientific status of the result.

The same distinction applies after merge: code or evidence may be canonical repository state while the physical interpretation remains only candidate, partial, negative, or UNKNOWN.

C.13 PDF acceptance

A release-quality textbook build requires more than a successful TeX exit code. The PDF must be rendered and inspected for clipping, overlapping text, broken glyphs, malformed tables and unresolved references. The page count, build log and hashes should be recorded with the release artifact. If the rendered document and the source disagree visually, the rendered document controls the acceptance decision.

A PDF is considered a current project snapshot only when both conditions hold: it builds from the declared repository state and the manuscript has been reconciled against the merged code/benchmark/card state it claims to describe.

Appendix D

Open Gates and Next Experiments

This appendix is deliberately a list of unresolved scientific gates rather than a roadmap written as if completion were guaranteed. The order below follows the current repository state on `main`; it is not a promise that every candidate will survive its next control.

D.1 Atomic correlation

1. Generalize the already informative one-coordinate carbon state-averaged active-external radial- p relaxation gate to a multidimensional state-averaged orbital optimizer/MCSCF-type control with simultaneous admissible active-external s , p , and d rotations. Experimental term energies remain excluded from the objective.
2. Establish basis/grid convergence of the state-averaged optimum rather than treating the best sampled one-coordinate point as a continuously converged solution.
3. Re-evaluate correlated spin-orbit structure only after the orbital-optimization layer is frozen.
4. Extend systematic correlation controls beyond the second period and separate core-valence from valence correlation.
5. Tighten the H-Kr control quality gate where high-precision comparison is intended; the present coarse virial gate is a coverage criterion, not a precision theorem.

D.2 Shell topology

1. Derive trajectory-like or current-density proxies from actual SCF/CI states rather than synthetic circular paths.
2. Test particle-hole invariants internally for $p^k \leftrightarrow p^{6-k}$ before consulting knot labels.
3. Test transfer of the same shell descriptors from $2p^k$ to $3p^k$ and, where data permit, $4p^k$.
4. Keep unrestricted 3D exchange permutation data separate from projected winding and effective-2D braid data.

D.3 Conformal knot candidate

1. Symbolically verify the numerically observed unit-circle roots of $V_K(q) - 1/2$ for the candidate singular family.
2. Freeze mirror/amphichirality metadata and a canonical knot-table normalization.
3. Run replacement-anchor controls for z_* and quantify the look-elsewhere effect.
4. Pre-register a shell-symmetry comparison rule before combining knot and shell labels.
5. Retain the existing failed direct spectral match as a negative control.

D.4 Compound and molecular state admission

The current molecular frontier is the incomplete v0.14A1 common relaxation screen, not a new fitted bond rule.

1. Complete only the missing ArBr_2 execution under the already frozen v0.14A1 B97M-V/VV10/def2-TZVPD, grid, SCF, optimizer and seed policy. A further timeout/cancellation/failure remains explicit evidence and does not license an ArBr_2 -specific rescue.
2. Freeze the complete 9/9 formula and 45/45 start ledger before opening a new physical admission layer. Original first-attempt receipts for the eight completed formulae remain identifiable and are not replaced by broader reruns.
3. Only after the full relaxation ledger exists, preregister v0.14B as a Hessian/local-minimum gate. Do not choose frequency thresholds, rescue branches or candidate exclusions after inspecting which states are convenient.
4. For states admitted as local minima, compare conventional energies under one common policy. Keep the lower-cost v0.14 screen distinct from the higher-level v0.8 energetic-admission contract.
5. Apply the v0.10 electronic-topology contract separately. A 3c4e/VDW label requires the declared independent diagnostic families and may remain UNKNOWN even when a stationary geometry exists.
6. Retain all competing v0.13 state branches that survive their declared gates; a composition may have multiple metastable states or none of the enumerated candidates.

D.5 Simple-system molecular controls

The $\text{H}_2^+/\text{H}_2/\text{HeH}^+/\text{LiH}$ ladder remains a required conventional validation track, although v0.14A/A1 means it is no longer accurate to call H_2^+ the repository’s first molecular implementation target.

1. Implement/replay H_2^+ as a one-electron two-centre control with all candidate topological/relational diagnostics disableable without changing the control solution.
2. Add H_2 and test exchange/correlation and density reorganization under the same control-first discipline.
3. Add HeH^+ and LiH as asymmetric and core–valence transfer tests.
4. Only after these simple controls and the present XY_2 admission work consider more complex multi-centre ring systems where loop language is visually compelling and therefore especially vulnerable to confirmation bias.

D.6 Atomic activation-control path

The v0.11/v0.12 atomic finite-difference route is not the next molecular classifier. Its numerical scans remain useful controls, but further diffuse-basis escalation is not launched automatically after the L3 krypton counterexample. Any renewed atomic descriptor programme must first define a new global numerical objective and prospective admission rule rather than tune the existing path to reproduce the molecular screen.

D.7 Relational/TIR/PhaseNav admission

The first admitted relational physics test must define a no-refit observable whose prediction is not already encoded in the control target. Until that exists, κ , W_{AB} , 36D encodings and affective

semantics remain metadata, representation machinery or candidate maps rather than corrections to validated Hamiltonians.

PhaseNav may provide the native command/gate/orchestration surface for new project functions. Conventional external electronic-structure packages, where used, remain isolated numerical control backends. A PhaseNav command or receipt may preserve provenance and gate logic, but it may not change a conventional energy, convergence failure or Hessian result to obtain a desired relational verdict.

D.8 Documentation and release gate

A scientific increment is not complete merely because code and benchmark files were merged. Before a release-quality PDF is called current:

1. reconcile `README.md` and `STATUS.md` with the merged `main` state;
2. update the relevant manuscript chapter, benchmark ledger, implementation map, validation contract and open-gate list;
3. run the maintained main/PR software, bibliography and textbook gates;
4. render-inspect the resulting PDF and record hashes/page count.

This prevents a build-correct but scientifically stale manuscript from being mistaken for the repository snapshot.

D.9 Promotion rule

An open gate is closed by a derivation, implementation and declared validation result, not by narrative plausibility. Any failure or partial result that changes the next hypothesis remains part of the permanent validation history. Repository merge and physical promotion remain separate decisions.

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Current textbook status

This document is a working research textbook reconciled against the repository state through the v0.14A1 molecular-screen checkpoint. Conventional atomic-control results retain their declared validation status; the shell-topology, conformal knot-shell and relational/TIR layers remain candidates unless explicitly stated otherwise. The first one-coordinate state-averaged carbon orbital-relaxation gate is variationally informative, with multidimensional state-averaged orbital optimization remaining open. The compound architecture now carries unranked competing relational states, and the common v0.14A1 molecular relaxation screen has durable first-attempt evidence for eight of nine formulae (40 of 45 frozen starts). ArBr_2 remains missing execution, no Hessian/local-minimum admission has yet been performed, and no 3c4e/VDW topology or validated ground-state ranking is inferred from the screen. The repository default branch `main` is the project Source of Truth; NOEMA/Library continuity artifacts are mirrors/provenance rather than prerequisites for reproducing this manuscript.